



THE ANNUAL MEETING OF THE ALASKA CHAPTER OF THE WILDLIFE SOCIETY



**Wood Student Center
University of Alaska, Fairbanks
Fairbanks, Alaska**

April 3-5, 2013



OFF-CAMPUS SITES

- 1 Robert G. White Large Animal Research Station, Yankovich Road
- 2 Harper Building, 4250 Geist Road (Interior-Aleutians Campus)
- 3 West Valley Plaza, 4001 Geist Road
- 4 Wells Fargo Building, 794 University Ave.
- 5 Facilities Services Division of Design and Construction, 590 University Ave.
- 6 Denali Building, 3352 College Road
- 7 Administrative Services Center, 3295 College Road
- 8 Central Receiving, Campus Mail Services, 1855 Marika Road
- 9 Polar Flat Research Range, 30 mile Steese Highway
- 10 Community and Technical College, 604 Barnette St.
- 11 Fire Station 2, 1950 University Ave. S.
- 12 West Side Business Park, 2175 University Ave. S. (Center for Distance Education, College of Rural and Community Development Bookstore, Statewide Office of Academic Affairs, Alaska Teacher Placement, Early Childhood Education, Math in a Cultural Context)
- 13 CTC Automotive Technology Center, 3202 Industrial Ave.
- 14 Fairbanks Pipeline Training Center, 2600 Cartwright Court
- 15 Tanana District Office of the Cooperative Extension Service, Fairbanks Community Food Bank building, 724 27th Ave.

STUDENT SERVICES

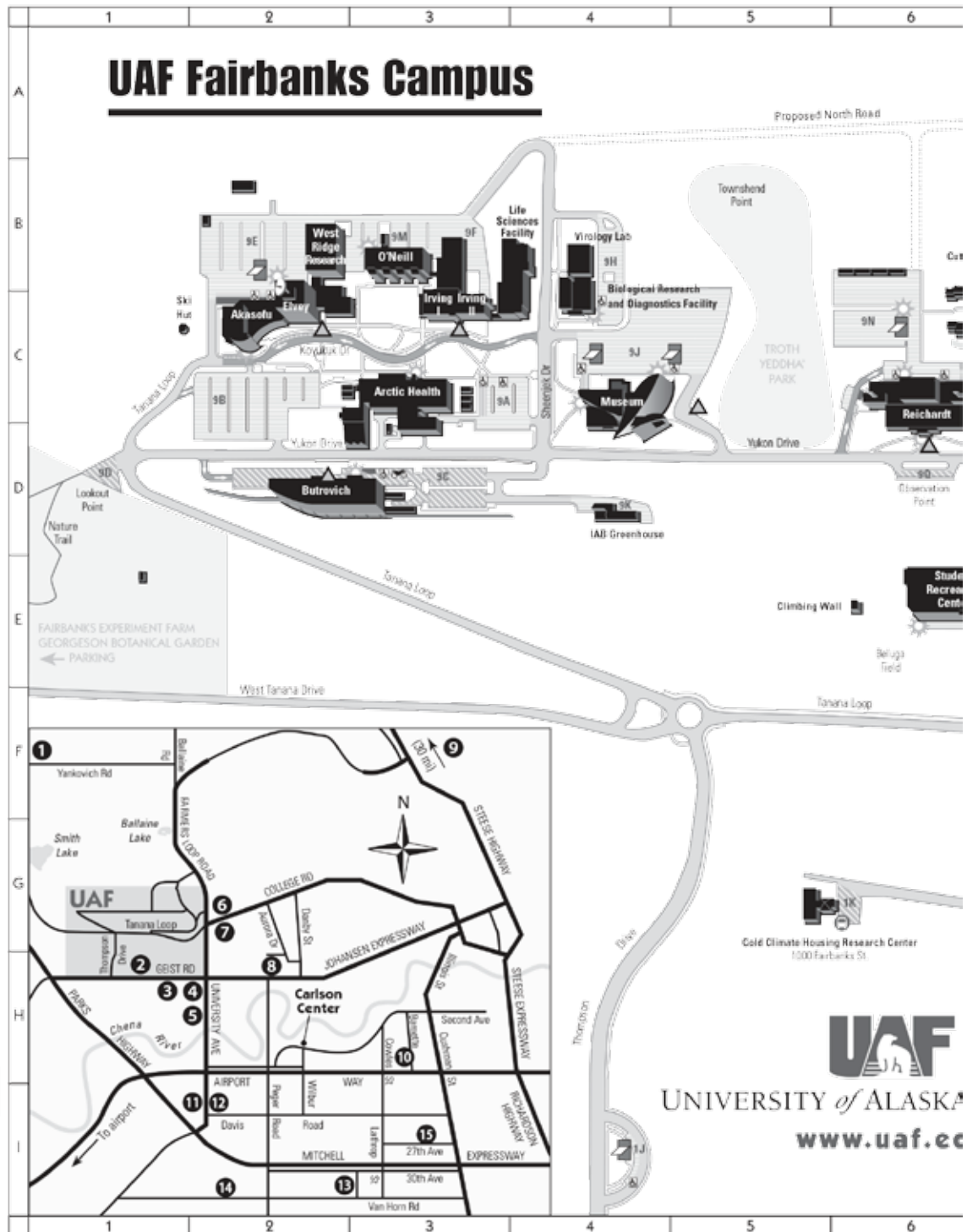
- Academic Advising (Gruening Bldg).....e-10
 Admissions (Signers' Hall).....e-10
 Business Office (Signers' Hall).....e-10
 Bookstore (Constitution Hall).....d-10
 Career Services (Eielson Bldg).....e-10
 Developmental Education (Eielson Bldg).....e-10
 Dining (Lola Tilly Commons).....f-9
 Disability Services (Whitaker Bldg).....d-10
 Financial Aid (Eielson Bldg).....e-10
 Graduate School (Eielson Bldg).....e-10
 Health & Counseling (Whitaker Bldg).....d-10
 Multicultural Affairs and Diversity (Eielson Bldg).....e-10
 New Student Orientation (Wood Center).....d-9/10
 Polar Express (Eielson Bldg).....e-10
 Registrar (Signers' Hall).....e-10
 Residence Life/Housing (M3S Complex).....e-8
 Rural Student Services (Brooks Bldg).....e-11
 Student Center (Wood Center).....e-9/10
 Student Recreation Center.....e-6
 Testing Services (Gruening Bldg).....e-10
 Veterans' Services (Eielson Bldg).....e-10
 Women's Center (Eielson Bldg).....e-10
 Writing Center (Gruening Bldg).....e-10

RESIDENCE HALLS AND HOUSING

- Bartlett Hall.....e-8
 Outlier Apartment Complex.....b/c-6/7
 Faculty Housing.....b-8, c-9
 Garden Apartments.....b-9
 Harwood Hall.....b-9
 Hess Village.....a/b-8
 Lathrop Hall.....e-8
 MacLean Hall.....b-9
 McIntosh Hall.....e-9
 Moore Hall.....c-7
 Nerland Hall.....e-9
 Nordic House.....b-9
 Skarland Hall.....e-8
 Stevens Hall.....e-8
 Stuart Hall.....c-8
 Walsh Hall.....e-10
 Wickersham Hall.....e-9

BUILDINGS

- Akasofu Bldg.....
 (International Arctic Research Center).....e-2
 Arctic Health Research Bldg.....e-3
 Atkinson Bldg (Power Plant).....g-10
 Biological Research and Diagnostics Facility.....b/c-4

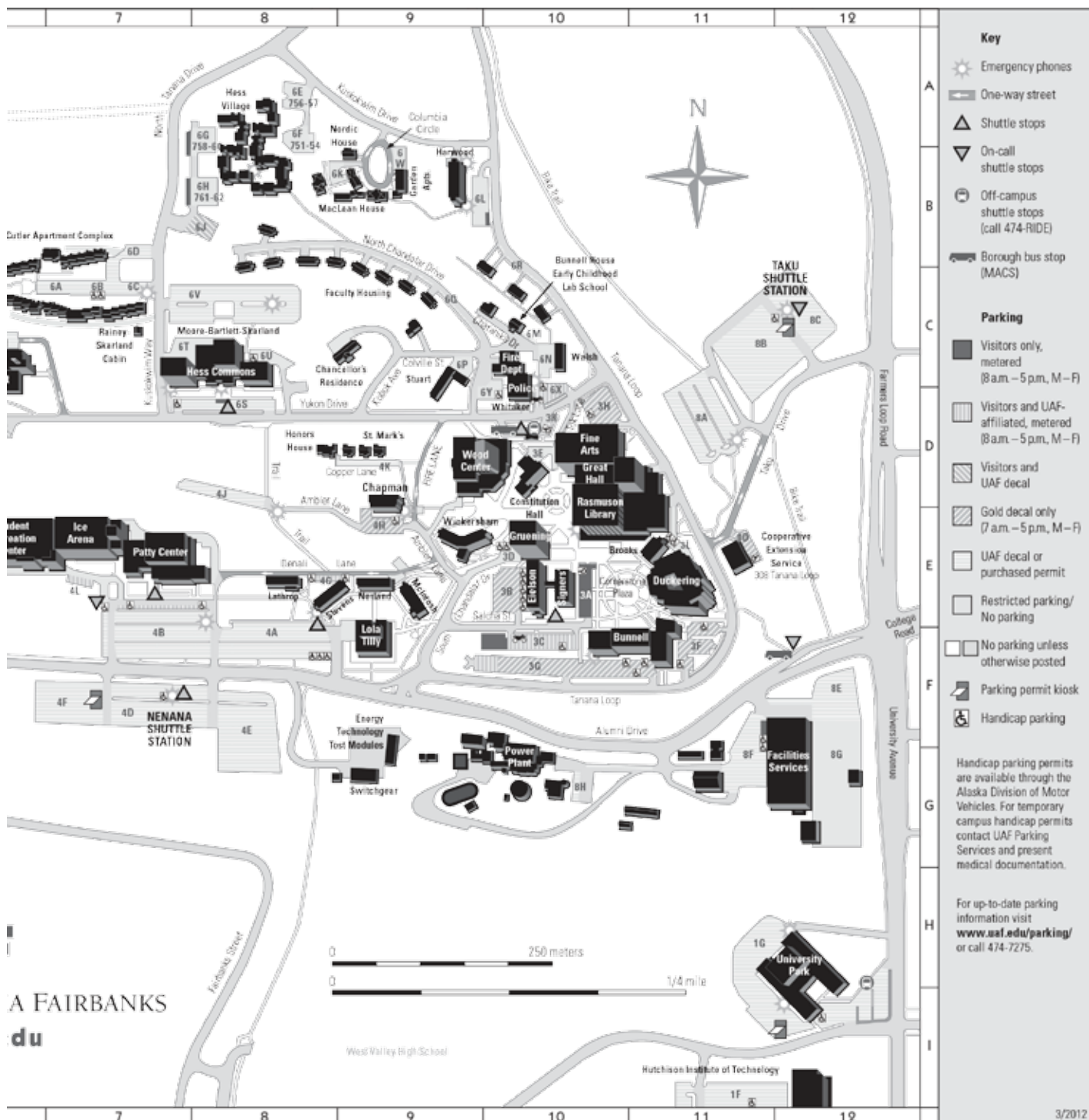


UAF
 UNIVERSITY of ALASKA
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PARKING LOTS

Name	code	grid
Administrative Services, East	1A	g-2
Administrative Services, West	1B	g-2
Arctic Health, East	9A	e-3
Arctic Health, West	9B	e-2
Ballaine	8A	d-11
Biological Research	9H	b/c-4
Bunnell, East	3F	f-11
Bunnell, South	3G	f-10/11
Bunnell House	6M	c-10
Butrovich	9C	d-2/3
Chapman	4H	d-9
Cold Climate Housing Research Center	1K	g-6
Constitution	3E	d-10
Cooperative Extension Service BD		e-11

Brooks Bldg	e-11	Fairbanks Experiment Farm	e-1	Salisbury Theatre (Great Hall)	d-10
Bunnell Bldg	f-10/11	Fine Arts/Theatre	d-10	Schaible Auditorium (Bunnell Bldg)	f-11
Butrovich Bldg	d-2/3	Great Hall, Regents'	d-10	Shuttle Station, Nenana	f-7
Chancellor's Residence	c-9	Greenhouse, IAB	d-4	Shuttle Station, Teku	c-12
Chapman Bldg	d-9	Gruening Bldg	e-10	Signers' Hall	e-10
Cold Climate Housing Research Center	g-5	Hess Commons	e-8	Ski Hut	e-1
Commons, Lola Tilly	f-9	Honors House	e-8	St. Mark's Catholic Center	e-9
Community and Technical College	h-3	Hutchinson Institute of Technology	i-12	Student Recreation Center	e-6
Constitution Hall	d-10	Ice Arena	e-7	University Park Bldg	h-12
Cooperative Extension Service Bldg	e-11	Irving I & II	e-3	Virology Lab, State of Alaska	b-4
Davis Concert Hall (Great Hall)	d-10	Life Sciences Facility	b-4	Wells Fargo Bldg	h-1/2
Duckering Bldg	e-11	Library, Elmer E. Rasmussen	d/e-10	West Ridge Research Bldg	b-2
Early Childhood Lab School (Bunnell House)	e-10	Museum of the North, UA	e-4	Whitaker Bldg	d-10
Eielson Bldg	e-10	O'Neill Bldg	b-3	Wood Center	d-9/10
Ewey Bldg	e-2	Patterson Center	e-7		
Facilities Services	1/g-12	Rainey-Skarland Cabin	c-7		
		Reichardt Bldg	e-6		



A FAIRBANKS
du

Copper Lane.....4K.....d-9	Lookout Point.....9D.....d-1	Physical Plant, West.....8F.....f/g-11
Cutler Apartment Complex...6A-D.....b/c-6/7	Lower Dorms.....4A.....f-8	Power Plant.....8H.....g-10
Denali Lane.....4G.....e-8/9	MacLean House.....6K.....b-8/9	Reichardt.....9N.....c-6
Eielson, South (meters).....3C.....f-10	MBS East.....6U.....c-8	Signers' (meters).....3A.....e-10
Eielson, West.....3B.....a-10	MBS North.....6V.....c-7/8	Shreejok.....9G.....b-4
Elvey.....9E.....b-2	MBS South.....6S.....d-8	Stuart.....6P.....c-9
Fine Arts.....3I.....d-10	MBS West.....6T.....c-7/8	Student Recreation Center.....4L.....e-7
Garden Apartments.....6W.....b-9	Museum of the North, UA.....9J.....c-4/5	Taku.....8C.....c-11/12
Greenhouse, UAF.....9K.....d-4	Nanana.....4D.....f-7	Taku, East.....8C.....c-11/12
Greening, West.....3D.....e-10	Nanana, East.....4E.....f-8	Tanana Apartments.....6R.....c-10
Haida.....4J.....d-8	Nanana, West.....4F.....f-7	Thompson Drive Pull-out.....1J.....f-4
Harper.....1E.....b-1	North Chandalar.....6O.....b/c-8/9	University Park.....1G/h-i-11/12
Harwood.....6L.....b-9	Observation Point.....9O.....d-6	Virology Lab, State of Alaska.....9H.....b/c-4
Hess Village.....6E-H.....a/b-8	O'Neill, North.....5M.....b-3	Walsh Hall.....6N.....c-10
Hutchison.....1F.....i-11	Patty Center/SRC.....4B.....f-7	Whitaker, East (meters).....6X.....c/d-10
Living.....9F.....b-3	Physical Plant, East.....8G.....f/g-12	Whitaker, West.....6Y.....c/d-10
Library, East.....3L.....e-11	Physical Plant, North.....8E.....f-12	Wood Center, East.....3K.....d-1

EMERGENCY PHONES

Administrative Services Center.....g-2	Patty Center Parking Lot.....e-8
Akassofa Bldg.....c-2	Rasmusen Library.....e-10
Arctic Health Research Bldg.....c-3	Reichardt Bldg (2).....c/d-6
Balaine Parking Lot.....d-11	Stevens Hall.....e-8
Bunnell Bldg.....f-10	Student Recreation Center.....e-6
Butovich Bldg.....d-3	University Park Parking Lot (2).....h/i-12
Chapman Bldg.....e-9	Whitaker Bldg.....c-10
Cutler Apartment Complex Parking Lot.....c-7	
Haida Parking Lot.....d/e-8	
Harwood Hall (2).....b-9	
Hess Commons.....c/d-8	
Hess Village Recreation Center.....b-8	
Moore Parking Lot.....c-7	
Moore-Bartlett-Skarland Complex.....c-8	
Museum of the North, UA (2).....c-4	
O'Neill Bldg.....b-3	

EMERGENCY SERVICES

Fire Station 1.....c-10	
Fire Station 2.....f-2	
Police.....c/d-10	

3/2012



Please thank our generous supporters.

ANNUAL MEETING OF THE ALASKA CHAPTER OF THE WILDLIFE SOCIETY

*WOOD STUDENT CENTER
UNIVERSITY OF ALASKA, FAIRBANKS
FAIRBANKS, ALASKA
APRIL 3-5, 2013*

CONFERENCE COMMITTEE

Program: Laura Prugh and Grant Hilderbrand

Local Arrangements: Kevin Colson, Kris Hundertmark, Jerry Hupp, Kaithryn Ott,
Marci Johnson and Travis Booms

Special Sessions and Workshops: Tom Paragi, Angela Romito, and Falk
Heuttmann

Registration and Support: Liz Solomon and Todd Rinaldi

Cover and Program Design: Mary Whalen

Cover Photos: Brian Uher-Koch, Chris Zimmerman, and Kyle Joly

Catering: UAF Catering

Conference-At-A-Glance

Wednesday, April 3

330 O'Neill - UAF

8:00 am - 12:00 pm Workshop: Machine Learning - Falk Huettmann, University of Alaska, Fairbanks

Wood Student Center - UAF

8:00 am - 12:00 pm Workshop: Structured Decision Making and Adaptive Management- Angela Romito, University of Georgia

Wood Student Center - UAF

7:30 am - 4:00 pm Registration

1:00 pm - 1:10 pm Welcome and opening remarks – Jerry Hupp, President

1:10 pm - 2:00 pm Plenary Session - Dr. Wayne Regelin, Retired, Alaska Department of Fish and Game

2:00 pm - 5:00 pm Contributed Oral Presentations

Wood Student Center

5:15 - 8:00 pm Student-Professional Mixer and Poster Session

Thursday, April 4

Wood Student Center - UAF

7:30 am - 4:00 pm Registration

8:00 am to 5:00 pm Special Session – Implications of managing fire succession and forest biomass productions on wildlife habitat

8:10 am to 11:50 am Fire Session

11:50 am to 1:20 pm, Lunch and Northwest Section Meeting

1:20 pm to 5:15 pm Forestry Session

5:15 pm to 8:00 pm Banquet and Keynote Address

Marvin Eng, M.Sc., RPBio – “The Mountain Pine Beetle in British Columbia: Reflections of a Deckhand During a Perfect Storm”

Friday, April 5

Wood Student Center - UAF

7:30 am - 12:00 pm Registration

8:00 am - 5:00 pm Contributed Oral Presentations

12:00 am - 1:40 pm Lunch and Alaska Chapter Business Meeting

Wednesday, April 3

University of Alaska Fairbanks

1:00 pm - 5:00 pm: Plenary Address and Contributed Oral Presentations

Wood Student Center, UAF

- 1:00 WELCOME AND OPENING REMARKS
Jerry Hupp, President
- 1:10 PLENARY ADDRESS
Dr. Wayne Regelin
- 2:00 FUTURE CLIMATE CHANGE IMPACTS ON BOREAL FOREST HABITAT FOR THREE RESIDENT BIRDS IN ALASKA
Jennifer Schmidt, Jacob LaCroix, Mark Olson, Thomas Kurkowski, and Scott Rupp
- 2:20 PREDICTING THE DISTRIBUTION OF HUMAN-MADE NOISE IN THE SOUNDSCAPE: AN INDICATOR OF HABITAT QUALITY
Timothy Mullet*, Falk Huettmann, Stuart Gage, and John Morton
- 2:40 BEHAVIORAL OBSERVATIONS OF BREEDING OSPREY IN INTERIOR ALASKA
Amal Ajmi
- 3:00 BREAK
- 3:20 MODELING AVIAN MIGRATION PHENOLOGY
Nicole Wells* and Mark Lindberg
- 3:40 BROOD MOVEMENT AND HABITAT USE BY AMERICAN AND PACIFIC GOLDEN-PLOVERS
Kelly Overduijn*, Colleen Handel, and Abby Powell
- 4:00 LAKE SHRINKAGE IN ALASKAN NATIONAL WILDLIFE REFUGES: HETEROGENEITY AND IMPLICATIONS
Brad Griffith, Jennifer Roach, and Vijay Patil
- 4:20 BREEDING ECOLOGY OF WHIMBRELS IN INTERIOR ALASKA
Christopher Harwood* and Abby Powell
- 4:40 BALD EAGLE NEST TURNOVER IN KENAI FJORDS NATIONAL PARK 2009-2012
Tammy Wilson and Laura Phillips
- 5:15 to 8:00 **Student-Professional Mixer and Poster Session**
Wood Student Center, UAF

Wednesday, April 3

Student-Professional Mixer and Poster Session

Wood Student Center, UAF

FEEDING BEHAVIOR OF CARIBOU (*RANGIFER TARANDUS GRANTI*) CALVES DOES NOT COMPENSATE FOR LOW FORAGE QUALITY

Perry Barboza and Rachelle Ruffner

DOG TICKS INTRODUCED AND ESTABLISHING IN ALASKA: INCREASED RISKS FOR TICK-BORNE ZOONOSSES

Kimberlee Beckmen

A BROADCAST SURVEY OF NORTHERN GOSHAWKS ON THREE RIVERS IN NORTHERN INTERIOR ALASKA

Tim Craig and Erica Craig

MOVEMENT AND HABITAT USE OF SNOWSHOE HARES IN INTERIOR ALASKA

Dashiell Feierabend*

A TOOL TO CREATE AND UPDATE A GEODATABASE WITH MONTHLY DATAFILES OF GPS LOCATIONS RECEIVED VIA ARGOS SATELLITES.

Cindy Hamfler and Jim Herriges

WOLF DENSITY AND TERRITORY SIZE IN A LOW BIOMASS UNGULATE PREY SYSTEM ESTIMATED WITH GPS COLLARS

Bryce Lake, Jason Caikoski, and Mark Bertram

NEW METHODS AND NUMBERS FOR DALL'S SHEEP MONITORING IN ALASKA'S NATIONAL PARKS

Kumi Rattenbury, Joshua Schmidt, Buck Mangipane, Brad Shults, Laura Phillips, Jim Lawler, and Maggie MacCluskie

ACTIVITY AND DIET OF LITTLE BROWN BATS (*MYOTIS LUCIFUGUS*) IN INTERIOR ALASKA

Rachel D. Shively*, Perry S. Barboza, Thomas S. Jung, and Craig S. Stricker

DIETARY ECOLOGY OF ALASKAN GRAY WOLVES: USING CAPTIVE WOLVES TO STUDY WILD POPULATIONS

Ashley Stanek*, Nathan Wolf, Shannon Jensen, Grant Hilderbrand, and Jeffrey Welker

VELERICORN ANTLERS IN ALASKAN MOOSE (*ALCES ALCES GIGAS*): THE ENDOCRINE DISRUPTION HYPOTHESIS

Raphaela Stimmelmayer and Julie Maier-Kitchen

MONITORING SUMMER FORAGES FOR ARCTIC CARIBOU: CORRELATING ISOTOPES WITH DIET AND LATITUDE

Lindsay VanSomeran*, Perry Barboza, David Gustine, Lincoln Parrett, and Craig Stricker

GEOPHAGY: A MEDIATOR OF PLANT SECONDARY COMPOUNDS IN SNOWSHOE HARES?

Suzanne Worker* and Knut Kielland

Thursday, April 4

University of Alaska Fairbanks

8:00 am - 5:00 pm: Implications of managing fire succession and forest biomass productions on wildlife habitat

Session Chair: Tom Paragi

8:00 WELCOME AND OPENING REMARKS
Tom Paragi

Fire Session

8:10 APPLICATIONS OF LONG TERM ECOLOGICAL RESEARCH FINDINGS IN ALASKA TO WILDLIFE MANAGEMENT
Knut Kielland

9:00 PERSISTENT EFFECTS OF FIRE SEVERITY ON EARLY SUCCESSIONAL FORESTS IN INTERIOR ALASKA
Aditi Shenoy*, Jill Johnstone, Eric Kasischke, and Knut Kielland

9:25 IMPACTS OF SHORTENED FIRE RETURN INTERVALS ON VEGETATION AND WILDLIFE HABITAT
Jennifer Barnes

9:50 BREAK

10:10 MANAGING MOOSE POPULATIONS FOR HARVEST IN WILDLAND FIRES
Kalin Kellie Seaton

10:35 CONSIDERATIONS IN MANAGING WILDLAND AND PRESCRIBED FIRE FOR RESOURCE OBJECTIVES
Tom Kurth

11:00 PANEL DISCUSSION – Fire Session

11:50 LUNCH and Northwest Section Meeting

Forestry Session

1:20 ASSESSMENT OF WOODY BIOMASS TO EVALUATE SUPPLY OPTIONS FOR HEAT AND POWER COGENERATION
Douglas Hanson*

1:45 CHALLENGES AND OPPORTUNITIES WITH BIOMASS REGENERATION FOR HARDWOODS AND SOFTWOODS IN THE BOREAL FOREST
Claire Doig

Thursday, April 4

- 2:10 CONSIDERATION OF SOIL NUTRIENT SUPPLY IN CHOOSING TREE HARVEST AND PROCESSING STRATEGIES
John Yarie
- 2:35 EFFECTS OF CLIMATE CHANGE ON ALASKA BOREAL FOREST GROWTH, HEALTH, AND REGENERATION
Glenn Juday, Claire Alix, and Tom Grant
- 3:00 BREAK
- 3:20 MODELING FOREST LANDSCAPE DYNAMICS TO EVALUATE MANAGEMENT OPTIONS
Nancy Fresco and T. Scott Rupp
- 3:45 LONG TERM CONTRACTS FOR TIMBER SUPPLY FROM STATE LANDS --- HOW WE GET THERE
Michael Curran
- 4:10 PANEL DISCUSSION – Forestry Session

5:15 pm - 8:00 pm: Banquet and Keynote Address

Marvin Eng, M.Sc., RPBio – “The Mountain Pine Beetle in British Columbia: Reflections of a Deckhand During a Perfect Storm”
Wood Student Center

Friday, April 5

University of Alaska Fairbanks

8:00 am - 5:00 pm: Contributed Oral Presentations

Wood Student Center, UAF

- 8:00 ASSESSING THE DISTRIBUTION AND GENETIC DIVERSITY OF THE ALASKAN HARE
Michelle Cason*, Link Olson, and Travis Booms
- 8:20 COMPARISON OF WILDLIFE RESOURCE SELECTION FUNCTION METHODOLOGIES: A CASE STUDY INVOLVING MOUNTAIN GOATS IN SOUTHEASTERN ALASKA
Dave Gregovich, Rod Flynn, and Kevin White
- 8:40 DISTRIBUTION AND MIGRATION OF AN IMPORTANT ARCTIC HERBIVORE, WITH IMPLICATIONS FOR SHRUB EXPANSION
Katie Christie*, Mark Lindberg, Roger Ruess, and Joel Schmutz
- 9:00 BROWSE INTAKES OF CARIBOU DURING SUMMER
Daniel Thompson* and Perry Barboza
- 9:20 DENSITY-DEPENDENT EFFECTS ON BODY CONDITION AND REPRODUCTION IN A COASTAL ALASKA MOOSE POPULATION
Kevin White, Neil Barten, Stacy Crouse, and John Crouse
- 9:40 MOOSE HUNTER SURVEY ON ANTLERLESS HUNTS
Todd Brinkman and Peter Fix
- 10:00 BREAK
- 10:20 LOW DENSITY WOLF AND MOOSE POPULATIONS ON THE YUKON FLATS OF ALASKA: INSIGHTS INTO PREDATION FROM THE PREDATORS' AND PREYS' PERSPECTIVE
Bryce Lake, Mark Bertram, Nikki Guldager, Jason Caikoski, and Robert Stephenson
- 10:40 RELATIONSHIP OF COMMUNITY CHARACTERISTICS TO HARVEST REPORTS BY ALASKAN MOOSE HUNTERS
Jennifer Schmidt, and Terry Chapin
- 11:00 RELATIONSHIPS BETWEEN MOVEMENT, DIET, AND HABITAT QUALITY OF URBAN MOOSE
Joseph Welch, Sean Farley, Perry Barboza, Don Spalinger, and David Battle
- 11:20 FINE-SCALE STRUCTURE OF CALVING LOCATIONS FROM NATAL PHILOPATRY IN MOOSE
Kevin Colson*, Kevin White, and Kris Hundertmark
- 11:40 DIETARY DIFFERENCES OF NORTHERN RED-BACKED VOLES (*MYODES RUTILUS*) FOUND ALONG AN ALASKAN MEGATRANSECT: A STABLE ISOTOPE ($\delta^{15}\text{N}$ AND $\delta^{13}\text{C}$) STUDY
Andy Baltensperger*, Falk Huettmann, and Jeffrey Welker
- 12:00 LUNCH and Alaska Chapter Business Meeting

- 1:40 THE EFFECT OF RESOURCE DEVELOPMENT ON ARCTIC FOX SURVIVAL ON THE ALASKAN ARCTIC COASTAL PLAIN
Garrett Savory*, Christine Hunter, Brian Person, Neil Lehner, and Nathan Pamperin
- 2:00 RELATIONSHIPS BETWEEN DIET, FECUNDITY, AND POPULATION ABUNDANCE AND COMPOSITION IN AMERICAN MARTEN IN INTERIOR ALASKA
Craig Gardner and Nathan Pamperin
- 2:20 SPATIAL PATTERNS ASSESSING POSITIVE VERSUS NEGATIVE PERCEPTIONS OF HUMAN-BEAR ENCOUNTERS
Kim Jochum*, Andrew Kliskey, Kris Hundertmark, Dmitry Lisitsyn, and Lillian Alessa
- 2:40 DOCUMENTING PRESENCE OF WOLVERINES IN NORTHEAST OREGON AND IDENTIFYING INDIVIDUALS
Audrey Magoun and Patrick Valkenburg
- 3:00 BREAK
- 3:20 POPULATION STRUCTURE OF TWO RABIES HOSTS IN ALASKA
Elizabeth Goldsmith*, Christopher Clements, Elisabeth Himschoot, Ben Renshaw, Kris Hundertmark, and Karsten Hueffer
- 3:40 SEXUAL SEGREGATION OF BROWN BEAR HABITAT USE ALONG THE SITUK RIVER
Anthony Crupi, Rodney Flynn, and LaVern Beier
- 4:00 ENHANCED RABIES SURVEILLANCE IN ALASKA USING THE DIRECT RAPID IMMUNOHISTOCHEMISTRY TEST (dRIT)
Stephanie Crawford, Kimberlee Beckmen, and David Sinnett
- 4:20 MODELING AND MAPPING FOREST DIVERSITY OF INTERIOR ALASKA AT 1-KM² RESOLUTION FOR CURRENT AND POSSIBLE FUTURE CLIMATE CONDITIONS
Brian Young, John Yarie, David Verbyla, Falk Huettmann, and F. Stuart Chapin
- 4:40 RESEARCH DESIGN FOR MANAGEMENT-RELATED WILDLIFE, HABITAT AND MONITORING FIELD STUDIES BASED ON MACHINE LEARNING ANALYSIS: NEW GOALS, OPPORTUNITIES AND WORKFLOWS
Falk Huettmann
- 5:00 BIRTH TIMING, SYNCHRONY, AND NEONATAL SURVIVAL IN A CRYPTIC FOREST UNGULATE: THE IMPORTANCE OF DETECTION METHOD FOR ESTIMATION
Sophie Gilbert, David Person, and Kris Hundertmark
- 5:20 ADJOURN

ABSTRACTS

SPECIAL SESSIONS AND CONTRIBUTED PAPERS AND POSTERS

Tuesday, April 3

1:00 pm

WELCOME AND OPENING REMARKS

Jerry Hupp

1:10 pm

PLENARY ADDRESS

Dr. Wayne Regelin, Retired, Alaska Department of Fish and Game

CONTRIBUTED ORAL PRESENTATIONS (2:00 pm - 5:00 pm)

2:00 pm

FUTURE CLIMATE CHANGE IMPACTS ON BOREAL FOREST HABITAT FOR THREE RESIDENT BIRDS IN ALASKA

Jennifer Schmidt¹, Jacob LaCroix, Mark Olson, Thomas Kurkowski², and Scott Rupp²

¹University of Alaska Fairbanks and ²University of Alaska Fairbanks. Contact: jischmidt@alaska.edu

As a result of projected climate change, fire and its impacts on boreal forests affect cavity-nesting birds which require old-growth habitat. We used a fire and vegetation succession model, ALFRESCO, with five down-scaled General Circulation Models (GCMs) and a moderate emission scenario (A1B), as an ensemble of future conditions, to evaluate the extent to which habitat, defined as old-growth (~80 years old) spruce forests, may be declining. We used simulated modeled outputs along with FRAGSTATS to examine habitat for boreal owls, three-toed woodpeckers and boreal chickadees over the next century. Chickadees and three-toed woodpeckers habitat decreased between 13-39% and boreal owl habitat decreased between 16-46% of current habitat. Losses were not uniform with some areas retaining or gaining habitat, like those around major river deltas. Under the HADLEY GCM, boreal forests could experience a mean reduction in black and white spruce cover and an increase in deciduous cover by 14% from 2007-2099. Simulated fire regime increased with all five GCMs compared to historic conditions. Inferring additional impacts, we hypothesize that boreal owls may be impacted by concentration of their predators as well as by reductions in their prey. Three-toed woodpeckers may experience a phenological mismatch with their prey. To conserve boreal chickadees and old-growth bird communities we recommend a hierarchical approach between local and regional landscape scales to conserve habitat and a greater awareness of climate change, metapopulation and community dynamics.

2:20 pm

PREDICTING THE DISTRIBUTION OF HUMAN-MADE NOISE IN THE SOUNDSCAPE: AN INDICATOR OF HABITAT QUALITY

Timothy Mullet^{1*}, Falk Huettmann², Stuart Gage³, and John Morton⁴

¹Biology and Wildlife Department, University of Alaska Fairbanks, ²EWHALE Lab, Biology and Wildlife Department, University of Alaska Fairbanks, ³Remote Environmental Assessment Laboratory, Michigan State University, and ⁴Kenai National Wildlife Refuge, U.S. Fish and Wildlife Service.

Contact: tcmullet@yahoo.com

The proliferation of mechanization over the last 200 years has contributed to the degradation of natural systems around the world. Sound is an important component of these natural systems and can be used as an indicator of habitat quality. A soundscape consists of the biological, anthropogenic, and geophysical sounds that emanate throughout a landscape. Soundscapes dominated by anthropogenic noise (anthrophony) may indicate that the quality of natural habitats has diminished. The Kenai National Wildlife Refuge is one of two refuges in Alaska on the road system and the only one with a significant urban interface. It manages 1.3 million acres of congressionally designated wilderness. Although wilderness areas are intended to preserve habitats in their natural state, the presence of anthrophony in these areas has become an increasing concern. In winter, wilderness areas are exposed to snowmachines, and air and road traffic. In order to determine what areas of the Refuge are affected by anthrophony we recorded sound data in wilderness and non-wilderness areas during the winter of 2010-2011. We quantified anthrophony using Power Spectral Density analysis and used machine learning software to predict anthrophony across the entire Refuge. Anthrophony was restricted to the northern wilderness areas where human activity is most prevalent. Our results show that these wilderness areas should be of conservation concern due to their level of anthropogenic noise. In the future, we intend to use this information to determine whether anthrophony effects moose stress levels and their spatial distribution.

2:40 pm

BEHAVIORAL OBSERVATIONS OF BREEDING OSPREY IN INTERIOR ALASKA

Amal Ajmi

ERM Alaska, Inc.. Contact: amal.ajmi@erm.com

This study provides the first account of breeding phenology and behavior for Osprey, *Pandion haliaetus carolinensis*, in Interior Alaska. Breeding Ospreys were observed at 2 nest sites on Fort Wainwright, Alaska. Breeding timelines and productivity were documented between 2006 and 2011. Detailed time-budget behavioral observations were conducted at the nest sites between 2008 and 2010 for adult Osprey (858 hours). Females incubated and brooded significantly more than male Osprey. Breeding Osprey behavioral reactions to human and nonhuman activities were also studied at Fort Wainwright, Alaska between 2009 and 2010. Ospreys were tolerant to aircraft and ground-based vehicular traffic; however were not as tolerant to human and avian activities. Female Ospreys were more reactive to human activities near the nest site, while males were more reactive to avian activities. Differences in behavioral reactions may be attributed to individual tolerance to activities and nest defense strategies.

3:20 pm

MODELING AVIAN MIGRATION PHENOLOGY

Nicole Wells^{1*} and Mark Lindberg²

¹US Fish and Wildlife Service and ²University of Alaska Fairbanks. Contact: nicole_torre@fws.gov

Changing global climate could result in temporal changes in phenological events for plant and animal species. Many birds have shifted their migration timing in response to warming temperatures, which could result in a timing mismatch with other ecological factors. Most previous studies of migration phenology did not account for detection probability, so timing of first arrival may therefore reflect time of first detection rather than actual arrival date. Furthermore, data are sparse during the time of first arrivals and monitoring a more data rich phase of migration phenology (i.e., peak arrival) may be more statistically viable and ecologically relevant. We assessed the ability to model migration phenology of birds while accounting for detection probability by analyzing data from the Tetlin National Wildlife Refuge, Alaska. We used a new occupancy modeling approach to estimate first arrival and peak arrival dates for American robins and mallards over a 10 year period. Comparisons of our results with traditional analysis methods suggest that not modeling detection in migration studies could result in bias and spurious results. Peak arrival is also shown to be a more ecologically meaningful and robust estimate than first arrival date.

3:40 pm

BROOD MOVEMENT AND HABITAT USE BY AMERICAN AND PACIFIC GOLDEN-PLOVERS

Kelly Overduijn^{1*}, Colleen Handel², and Abby Powell³

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The Seward Peninsula, Alaska, is an important sympatric breeding area for American Golden-Plovers (*Pluvialis dominica*) and Pacific Golden-Plovers (*P. fulva*). These species differ in their breeding ecology; Americans generally nest in xeric sites on higher slopes with little vegetation, whereas Pacifics generally nest in mesic areas with dense vegetation. Though many aspects of their biology have been well studied, little is known about brood-rearing habitat and brood movements of either species. During summer 2012, we located nests (American n = 25, Pacific n = 16), monitored nest success, and followed broods from hatch through fledge. We found more overlap between nesting and brood-rearing sites than expected. Additionally, some Americans moved broods to densely vegetated, mesic sites and some Pacifics moved broods to rocky, xeric sites on the top of ridges. Apparent survival of broods was 88%; with both adults attending chicks through fledge. Most broods traveled < 1 km from their nest sites to brood-rearing habitat. This information, along with vegetation mapping and monitoring food resources within nesting and brood-rearing habitats, will help us better understand why broods move and project potential effects of climate change on their breeding ecology.

4:00 pm

LAKE SHRINKAGE IN ALASKAN NATIONAL WILDLIFE REFUGES: HETEROGENEITY AND IMPLICATIONS

Brad Griffith¹, Jennifer Roach², and Vijay Patil³

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We used remote sensing and field studies to estimate the rates of change in lake size among 10 study areas in eight National Wildlife Refuges in Alaska over the past ~25 years and to identify mechanisms associated with lake size trends. Among all study areas there was a net decline of 0.8%/yr in lake size, but study area rates of change ranged from +0.34% to -2.96%/yr. Compounded annual rates suggest that average lake size for five of 10 study areas could decline by 33-77% over 50 years. Contrasts between populations of lakes with significant increasing or decreasing trends revealed that drying lakes were more likely to have wide floating vegetation mats, more likely to have burned, more likely to be on coarse-grained soils and were farther from rivers and streams. If current trends continue, there may be substantial effects on the abundance and distribution of both aquatic and terrestrial wildlife habitats and associated species richness. Preliminary results from the analysis of biological data from ~130 lakes in Yukon Flats NWR suggest that continued lake drying could ultimately result in increases in forest habitat at the expense of grass, sedge and shrub communities with negative implications for waterfowl and moose. Lake size, amount of wetland vegetation, proximity to rivers, and lake sizes within 5 km were the best preliminary predictors of waterfowl species richness. Projections of future lake sizes, assuming current trends continue, may be used to identify lakes that will retain or lose waterfowl species richness in a warming climate.

4:20 pm

BREEDING ECOLOGY OF WHIMBRELS IN INTERIOR ALASKA

Christopher Harwood^{1*} and Abby Powell²

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Despite the Whimbrel's widespread designation as a species of conservation concern, studies of its breeding ecology in North America, and especially Alaska, are few and geographically restricted. Known breeding locations within interior Alaska are especially patchily distributed. We have investigated the breeding ecology of a small (ca. 10-20 pairs annually) Whimbrel population at Kanuti NWR in northern interior Alaska since 2008; a second similarly sized population 15 km away was only discovered in 2011, despite helicopter surveys there the previous two years. Both populations show clumped nest dispersion, with nearest neighbor distances as close as 60 m, possibly the closest documented for the species. Initiation (and consequentially hatch) tends to be synchronous (4-6 d). Initiation in 2012 spanned ~ 13 days, suggesting re-nesting and/or late nesting occurred. Nest survival (assuming a constant daily survival rate and 28-d exposure) ranged from 20-60% from 2009-2012, although with marked disparity noted between the two populations in 2011-2012. High site fidelity is likely; 80% of marked males and 56% of marked females observed in 2011 were re-sighted in 2012. Identification of additional breeding areas in interior Alaska is planned for spring 2013.

4:40 pm

BALD EAGLE NEST TURNOVER IN KENAI FJORDS NATIONAL PARK 2009-2012

Tammy Wilson and Laura Phillips

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The use of indirect indicators of activity to monitor animal populations is becoming increasingly more common. Observations of animal use at persistent structures, such as nests, are among the most common of these monitoring techniques. In this case, the location of individual structures is known, while animal activity is observed as one of at least two states (e.g. presence/absence, reproduction success/failure, etc.). The probability that the structure survives from one year to the next, and the between-year dynamics of use both provide information about the health of the animal population. We used repeated annual surveys of bald eagle nests to evaluate inter-annual variation in eagle activity at nests in Kenai Fjords National Park. During helicopter-based nest surveys in May 2009-2012, we observed bald eagle nest to be "active". If an adult eagle was seen in the nest sitting in a posture consistent with egg incubation, or empty otherwise. Because an empty nest cannot be observed without error, we used a multi-state capture-recapture model to estimate nest turnover probabilities and nest survival. We found that temporary inactivity at nests was common for bald eagles in Kenai Fjords. The probability of nests transitioning from active to inactive, and from inactive to active between years was 0.33 and 0.34, respectively. We estimated that the physical structure of bald eagle nests "survived" for approximately 99 years, but that our survival estimate may be over-optimistic, because, all but the simplest model failed to converge using our four-year data set. Data from additional surveys will be necessary to improve confidence of nest transition and survival estimates. Our ability to model activity at nests and nest survival would be improved by implementing a Pollock's Robust Design, which would require an additional sampling period in May.

Tuesday, April 3

Student-Professional Mixer and Poster Session (5:15 pm - 8:00 pm)

FEEDING BEHAVIOR OF CARIBOU (*RANGIFER TARANDUS GRANTI*) CALVES DOES NOT COMPENSATE FOR LOW FORAGE QUALITY

Perry Barboza and Rachelle Ruffner

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We studied 7 male and 7 female calves to monitor consumption of milk and solid food from birth to 14 weeks of age in 1.5 ha wooded pens that were 37% pasture. Mothers and calves were provided with pelleted feeds to vary digestible energy (16 vs. 12 kJ/g) and protein (15.2 vs. 7.6%) that simulated the quality of summer forages: high energy and protein (HEHP = browse), low energy and protein (LELP = senescent graminoids) and low energy high protein (LEHP = emerging graminoids similar to the pasture). We observed the behavior of each calf in the morning (8am), afternoon (2pm) and evening (8 pm) of each day for a total of 3,958 records. Observations of suckling declined (18% - 3%) as those of grazing inclined (18% to 3%) from birth to 6 weeks of age. Calves also increased feeding on pelleted food to 4 weeks of age, which coincides with the development of ruminal function. Spacing between animals increased from morning to evening as calves were observed to feed more frequently in the evening from 4 to 14 weeks of age. Growth was slowest on the LELP diet (0.28 g/d) and fastest on the HEHP diet (0.48 g/d). Feeding was not affected by diet treatment, which suggests that calves were not able to compensate for low food quality by foraging or suckling more frequently. Caribou are most vulnerable to declines in diet quality after 4 weeks of age when suckling frequency declines and growth must be supported by solid food through late summer.

DOG TICKS INTRODUCED AND ESTABLISHING IN ALASKA: INCREASED RISKS FOR TICK-BORNE ZONOTIC DISEASES

Kimberlee Beckmen

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Ticks on small mammals and birds are endemic to Alaska however; dog ticks were not thought to be present. In 18 months, we detected numerous "dog" ticks which are also potential vectors of zoonotic diseases. The ADF&G tick surveillance program was established primarily for the early detection of the Moose Winter Tick (*Dermacentor albipictus*) which is already establishing in the Yukon on elk and moose. We received 28 ticks (identifications by Dr. Lance Durden), of which 18 were from dogs, 1 each from a cat, hare, marten, and 6 on people. Nine were American dog ticks (*Demacentor variabilis*) including from a dog that never left Juneau, and a dog that picked up the tick at Potter's Marsh in Anchorage. The dogs carrying *D. variabilis* resided in many areas of the state including Fairbanks and North Pole. *D. variabilis* is a vector of Rocky Mountain Spotted Fever (RMSF) and tularemia. There were three Brown dog ticks (*Rhipicephalus sanguineus*) on resident dogs of Sitka and Anchorage. This tick is a vector of canine ehrlichiosis, canine babesiosis, Lyme Disease, RMSF, & Q fever. Two Rocky Mountain Wood ticks (*Dermacentor andersoni*) were identified, which are vector RMSF, Anaplasmosis, tick paralysis, tick fever and Q fever. One Lone star tick (*Amblyomma americanum*) was identified. This tick, a vector of Ehrlichiosis and tularemia, was found on a dog in Eagle River. Our goal is to increase awareness about prevention necessary for pets traveling to Alaska and for practitioners to consider tick-borne diseases in their differential diagnoses.

A BROADCAST SURVEY OF NORTHERN GOSHAWKS ON THREE RIVERS IN NORTHERN INTERIOR ALASKA

Tim Craig¹ and Erica Craig²

¹Kanuti National Wildlife Refuge, US Fish and Wildlife Service and ²Aquila Environmental.

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We surveyed Northern Goshawks (*Accipiter gentilis*) on three rivers on Kanuti National Wildlife Refuge (KNWR) in 2011 and 2012 using a protocol slightly modified from that developed by the US Forest Service in the coterminous states. To our knowledge, this is the first systematic survey of Northern Goshawks that has taken place in the North American Arctic. We broadcast goshawk alarm calls at 79 stations over 45 km of the Kanuti River in 2011 and 2012, and 164 stations over 90 km of the Jim and South Fork Koyukuk Rivers in 2012. Response rates were low, with four and six responses on Kanuti River in 2011 and 2012, respectively, and five responses on Jim and South Fork Koyukuk Rivers in 2012. Because of the close proximity of several of the stations where goshawks responded, we suspect some of these birds were from the same territories. We elicited responses from species other than goshawks during our survey, as well. Eight different species responded at 41 stations on Kanuti River, with American Robins, red squirrels and Gray Jays the most frequent respondents. We elicited responses from 16 other species at 102 stations on the Jim and South Fork Koyukuk River route with Spotted Sandpipers, red squirrels and Gray Jays responding most frequently.

MOVEMENT AND HABITAT USE OF SNOWSHOE HARES IN INTERIOR ALASKA

Dashiell Feierabend*

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Snowshoe hares (*Lepus americanus*) are generally sedentary, but are known to use different habitats in summer. Hares in the boreal forest may also move among suitable habitats on a shorter time scale to maximize use of preferred foods and cover from predators. We used locations of GPS-collared hares from two characteristic boreal habitats to estimate seasonal home range size, habitat use, and diel patterns of movement among and within habitats. Seasonal home ranges were marginally larger in a black spruce (*Picea mariana*) forest during winter than in an early successional forest during summer. Hares captured in black spruce forest in late winter had multiple core use areas that regularly included white spruce (*P. glauca*)-birch (*Betula alaskana*) forest. Hares captured in an early successional forest in summer spent the majority of time in that habitat but also regularly used white spruce and black spruce forest. Movement rates were highest around midnight and lowest in mid-afternoon with at least a 4-fold difference between peak and nadir. The majority of movement between habitat types was coincident with times of peak movement rates. Our findings indicate that hares living in a mosaic of vegetation communities often concentrate their habitat use in multiple vegetation types even when food and cover are apparently abundant in a single habitat. Hares move between these areas on a daily basis, probably to make use of better foraging opportunities in one location and return to resting sites located in dense cover in a different location.

A TOOL TO CREATE AND UPDATE A GEODATABASE WITH MONTHLY DATAFILES OF GPS LOCATIONS RECEIVED VIA ARGOS SATELLITES

Cindy Hamfler¹ and Jim Herriges²

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We developed a tool to append monthly sets of Telonics GPS radiocollar data (transmitted through ARGOS satellite) into a master ESRI file geodatabase containing all point locations since collar deployment. Monthly data files from CLS America are manipulated in three steps with Telonics Data Converter software, an Excel macro, and an ArcGIS model. We hope to share this tool as well as learn of others'; experience in converting GPS data to geodatabases.

WOLF DENSITY AND TERRITORY SIZE IN A LOW BIOMASS UNGULATE PREY SYSTEM ESTIMATED WITH GPS COLLARS

Bryce Lake¹, Jason Caikoski², and Mark Bertram¹

¹US Fish and Wildlife Service - Yukon Flats National Wildlife Refuge and ²Alaska Department of Fish and Game. Contact: Bryce_Lake@fws.gov

Population density is a fundamental parameter for wildlife management. We estimated the density and territory size of wolves during 2009-2010 with locations acquired from global positioning system (GPS) collars. We conducted our study on the Yukon Flats of eastern interior Alaska, a region where moose were the sole ungulate prey and occurred at low densities and biomass. We compared between winter (n=5 packs) and annual (n=2 packs) territory size and between minimum convex polygon and adaptive kernel methods. Pack sizes averaged five in November 2009 and 4.8 in March 2010. Average winter territory size for five packs was 1,378 km² with 95% adaptive kernel and 1,421 km² with minimum convex polygon. Annual territory size for two packs averaged 1,395 km² with 95% adaptive kernel and 1,515 km² with minimum convex polygon. Density was 3.7 wolves 1000 km⁻² in fall and 3.5 in spring with 95% adaptive kernel, and 3.6 wolves 1000 km⁻² in fall and 3.5 in spring with minimum convex polygon. Territories were large and differed by 3-8% between methods used to analyze the locations. Winter territory size described 85-99% of annual territory size for two packs. Densities were low, not sensitive to the method used to analyze the locations, and consistent with the low biomass ungulate prey system on the Yukon Flats. Wolf harvest on the Yukon Flats was light in most years (0.46 wolves harvested 1000 km⁻², 21 wolves 45,731 km⁻²), and wolves were only occasionally moderately harvested.

NEW METHODS AND NUMBERS FOR DALL'S SHEEP MONITORING IN ALASKA'S NATIONAL PARKS

Kumi Rattenbury, Joshua Schmidt, Buck Mangipane, Brad Shults, Laura Phillips, Jim Lawler, and Maggie MacCluskie

National Park Service. Contact: kumi_rattenbury@nps.gov

Over 76,000 km² (29,300 mi²) of potential Dall's sheep habitat is in national parks in Alaska, and the last comprehensive surveys counted 30,455 total sheep, or as much as 40% of Alaska's sheep, in these parks in the early 1980s (Singer 1984). In 2009, we began testing aerial distance sampling survey methods coupled with Bayesian hierarchical modeling to update abundance and composition estimates with measures of precision for Dall's sheep in Denali, Gates of the Arctic, Lake Clark, Noatak, and Wrangell-St. Elias National Parks and Preserves. By borrowing detection information and informed priors from multiple surveys for the Bayesian process, we estimated total abundance, ewe-likes (ewes, yearlings and small rams), lambs, < full-curl rams, and ratios of lambs per 100 ewe-like and rams per 100 ewe-like with CV's $\leq 15\%$, even for smaller areas such as the western Baird Mountains of Noatak (1,842 km², 711 mi²) and the Iktalik preserve subarea of Gates of the Arctic (2,542 km², 981 mi²). Estimates for $\geq$ full-curl rams were less precise because of low numbers of detection, though we can improve upon this with repeated surveys. We estimate that approximately 27-28,000 sheep were in these parks from 2010-2012. These methods will be incorporated into long-term monitoring programs to inform harvest management and promote conservation in these parks.

ACTIVITY AND DIET OF LITTLE BROWN BATS (*MYOTIS LUCIFUGUS*) IN INTERIOR ALASKA

Rachel D. Shively^{*1}, Perry S. Barboza¹, Thomas S. Jung², and Craig S. Stricker³

¹Institute of Arctic Biology, Department of Biology and Wildlife, University of Alaska, Fairbanks, ²Fish and Wildlife Branch, Environment Yukon, and ³U. S. Geological Survey, Fort Collins Science Center. Contact: rdshively@alaska.edu

Little brown bats (*Myotis lucifugus*) are becoming more common in Alaska due to climate change, but the diet, range, migration routes and winter hibernacula are currently unknown. Moreover, the species is threatened by the westward advance of White Nose Syndrome. This project combines citizen science with wildlife tracking to assess the distribution and adaptability of *M. lucifugus* in interior Alaska. Interviewing individuals throughout the state has provided reports of bats as far north as Wainwright (70.647185, -160.016216). Nightly foraging times at one roost in Moose Creek (64.712474, -147.161128) increased from 190 min (00:11 - 02:59) on July 17th to 500 min on (22:44 - 07:04) on August 20th when migration began. We captured 23 bats at Moose Creek before migration from 9 to 13 August 2012. Estimated fat content of bats varied widely from 0.86 to 2.71g with 11 animals below 2g, which is the fat reserve required to survive hibernation. Captures also included 2 pregnant females that would have been unlikely to survive hibernation. Hair analysis of captured bats provided values for $\delta^{15}\text{N}$ from 5.68 to 8.17‰, $\delta^{13}\text{C}$ -28.45 to -24.45‰, and $\delta^{34}\text{S}$ 2.7 to 5.5‰ which were similar to values of $\delta^{15}\text{N}$ from 7.50 to 10.94‰, $\delta^{13}\text{C}$ from -28.61 to -24.10‰, and $\delta^{34}\text{S}$ from 4.6 to 12.1‰ from bats collected in SE Alaska. Ongoing analyses of fecal fragments will be used to identify diet for comparison with isotopic values of hair.

DIETARY ECOLOGY OF ALASKAN GRAY WOLVES: USING CAPTIVE WOLVES TO STUDY WILD POPULATIONS

Ashley Stanek^{*1}, Nathan Wolf², Shannon Jensen³, Grant Hilderbrand⁴, and Jeffrey Welker¹

¹Department of Biological Sciences, University of Alaska Anchorage, ²Environment and Natural Resources Institute, University of Alaska Anchorage, ³The Alaska Zoo, and ⁴Alaska Regional Office, National Park Service. Contact: aestanek@alaska.edu

The effective use of carbon (¹³C) and nitrogen (¹⁵N) stable isotopes to study animal dietary ecology requires that we understand how quickly and how faithfully the isotopic values of resources are incorporated into the consumer tissues in which we are interested. These parameters, referred to as the incorporation rate and diet to tissue discrimination value, can vary between tissues, species, and diets. Consequently, it is essential that we use incorporation rates and diet to tissue discrimination values that are appropriate for our study. The National Park Service (NPS) is currently using stable isotope analysis to study the diets of wolves (*Canis lupus*) in Lake Clark National Park (Alaska, USA). Here, we present the results of a diet switch experiment designed to augment the NPS study by measuring the carbon and nitrogen stable isotope incorporation rates and diet to tissue discrimination values for plasma, serum, red blood cells, blood clots, hair and vibrissae of six adult wolves housed at the Alaska Zoo (Anchorage, Alaska, USA). Our results indicate that tissue specific discrimination values for wolves on a marine salmon diet ranged between 1.77% to 6.05% for carbon and -0.49% to 3.68% for nitrogen. In addition, we found that wolf plasma and serum can be used to infer diet over the previous 2-3 weeks, and that the carbon and nitrogen stable isotope values of wolf red blood cells and blood clots are representative of dietary items consumed over a period longer than 70 days.

VELERICORN ANTLERS IN ALASKAN MOOSE (*ALCES ALCES GIGAS*): THE ENDOCRINE DISRUPTION HYPOTHESIS

Raphaela Stimmelmayer¹ and Julie Maier-Kitchen²

¹Department of Wildlife Management North Slope Borough and ²Institute of Arctic Biology, University of Alaska Fairbanks. Contact: raphaela.stimmelmayer@north-slope.org

Feces are used to determine diet composition of herbivores by counting indigestible fragments (microhistology). Chemical markers may improve determination of highly digestible forages such as lichen that are difficult to identify by microhistology. Perlatolic acid (PA) is a phenolic toxin produced by *Cladonia* and *Cladonia* lichens that are favored by caribou and reindeer. We tested the hypothesis that concentrations of perlatolic acid in feces would reflect lichen consumption in reindeer and caribou. Captive reindeer were fed a complete diet free of PA with only 19 ppm phenols before switching to 100% *Cladonia stellaris* that contained 1137 ppm phenol with 51 units of PA. Fecal concentration of PA increased from 0 to 0.8 - 1.1 units one day after lichen feeding and remained at 0.1 - 1.6 units on days 3 and 5. However, PA concentrations varied from 0 to 1.7 units in feces from wild caribou that were estimated to consume 72-76% lichen by microhistology. Similarly, total phenols ranged from 92 to 722 ppm in wild caribou consuming 60 -76% lichen. Although PA quickly reflected lichen consumption in captive animals, large variations within animals between days probably reflect changes in flow and food intake. High variation in PA and phenols of wild caribou probably reflect variations in intake and flow as well as regional and specific differences in phenolic concentrations of lichens. Lichen toxins may be used as a qualitative tool to assess selection of specific lichens but are not suitable as a marker for intake of lichen.

MONITORING SUMMER FORAGES FOR ARCTIC CARIBOU: CORRELATING ISOTOPES WITH DIET AND LATITUDE

Lindsay VanSomerén^{*1}, Perry Barboza¹, David Gustine³, Lincoln Parrett⁴, and Craig Stricker⁵

¹Institute of Arctic Biology, Department of Biology and Wildlife, University of Alaska Fairbanks, ³U. S. Geological Survey, Alaska Science Center, ⁴Alaska Department of Fish and Game, and ⁵U. S. Geological Survey, Fort Collins Science Center. Contact: lindsayvansomerén@gmail.com

Isotopic markers in feces and tissues are a potential tool for monitoring the importance of feeding areas for migratory herbivores such as caribou. We collected six species of preferred forages along a latitudinal gradient in the summer ranges of the Central Arctic (9 sites) and Teshekpuk Lake (4 sites) caribou herds during 2011. We analyzed forages at peak protein content to compare maximum diet quality and isotopic markers among species and sites. Dicots (*Salix pulchra*, *S. richardsonii*, and *Pedicularis langsдорфii*) contained more protein ($21.0\% \pm 3.3\%$ vs. $16.5\% \pm 2.8\%$; $p < 0.01$) and digestible dry matter ($79.2\% \pm 11.3\%$ vs. $56.6\% \pm 4.8\%$; $p < 0.01$) than monocots (*Carex aquatilis*, *C. bigelowii*, and *Eriophorum vaginatum*). The range of values for isotopes of nitrogen ($\delta^{15}\text{N}$ -6.6 – +4.3), and sulfur ($\delta^{34}\text{S}$ -3.6 – +15.5) were greater than those for carbon ($\delta^{13}\text{C}$ -29.8 – -24.9). Latitude did not affect $\delta^{15}\text{N}$ of forages, while $\delta^{34}\text{S}$ may indicate coastal habitats: $\delta^{34}\text{S}$ was higher on the coastal plain than in the foothills for *S. pulchra* (11.1 ± 3.6 and 3.1 ± 2.6 ; $P < 0.01$). Small declines in $\delta^{13}\text{C}$ with latitude for *E. vaginatum* ($P = 0.01$), *P. langsдорфii* ($P < 0.01$), *S. pulchra* ($P < 0.01$), and *S. richardsonii* ($P = 0.03$) were probably associated with changes in water availability from the foothills to the coast. Isotopic N may broadly reflect the relative importance of high quality dicots to caribou because monocots were higher than dicots in $\delta^{15}\text{N}$ (2.1 ± 1.1 vs. -3.0 ± 2.3 ; $P < 0.01$).

GEOPHAGY: A MEDIATOR OF PLANT SECONDARY COMPOUNDS IN SNOWSHOE HARES?

Suzanne Worker^{*} and Knut Kielland

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Consumption of mineral soil by herbivores is thought to have several physiological benefits, including the mediation of plant secondary metabolites (PSMs). Using soil from a known lick in northern Alaska, we investigated the effect of geophagy on feed intake, weight maintenance and nutrient balance of captive snowshoe hares (*Lepus americanus*) fed a formulated ration or willow leaves (*Salix alaxensis*). We also compared feed intake and weight loss in hares offered lick soil or a calcium supplement (calcium carbonate or calcium chloride) and fed *S. alaxensis* twigs. For hares fed willow leaves, those with access to soil lost less weight and had higher daily feed intake and higher digestible nitrogen, calcium, magnesium, sodium, and potassium intakes compared to hares without access to soil. Hares offered calcium carbonate had higher feed intake than hares offered soil, though weight loss was not significantly different. Calcium chloride supplements were used only sparingly and animals on this treatment ate less and lost more weight than hares offered calcium carbonate or soil. It appears that ingestion of lick soil is an effective strategy in mediating the effects of tannins in natural browse. While soils implicated in the mediation of PSMs are often very high in clay, in this case, the mineral constituents of soil apparently have an important role.

Thursday, April 4

IMPLICATIONS OF MANAGING FIRE SUCCESSION AND FOREST BIOMASS PRODUCTIONS ON WILDLIFE HABITAT (8:00 am - 5:00 pm)

Session Chair: Tom Paragi

Fire Session

8:10 am

APPLICATIONS OF LONG TERM ECOLOGICAL RESEARCH FINDINGS IN ALASKA TO WILDLIFE MANAGEMENT

Knut Kielland

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The Bonanza Creek Long Term Ecological Research (BNZ LTER) program, started in 1987, was initially focused on studying successional dynamics of floodplain and upland ecosystems. More recently the scope of these investigations has fallen under the umbrella of changing disturbance regimes modulated by climate change, and the important consequences of these ecosystems changes to social-ecological systems. During this program evolution, studies of the interactions between forest dynamics and wildlife, especially mammalian herbivory, have remained a cornerstone of BNZ LTER research. Here I review some of the findings derived from long-term observations and experiments, pertaining to plant and animal populations, of relevance to forestry and wildlife professionals. My first aim is to review some of the complex, but not incomprehensible, interactions between soils, plants and wildlife which shape the boreal forest of interior Alaska. Sound management of this landscape is predicated on an understanding of these processes. My second aim is to illustrate how cooperation between the University of Alaska and resource specialists within state and federal agencies promotes common goals, and to call for increased collaborations between university researchers, students and managers to address the challenges of sustaining our shared natural resources.

9:00 am

PERSISTENT EFFECTS OF FIRE SEVERITY ON EARLY SUCCESSIONAL FORESTS IN INTERIOR AK.

Aditi Shenoy^{*1}, Jill Johnstone², Eric Kasischke³, and Knut Kielland⁴

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In recent decades increases in fire frequency, size, and severity have been observed in black spruce forests of interior Alaska. The extent to which the forest floor organic layer is consumed during fire affects patterns of germination and stand development. Severe burning which removes the organic layer, promotes the recruitment of deciduous tree species, such as aspen, while decreasing the relative abundance of black spruce immediately after fire. By contrast, light burning favors self-replacement of black spruce. In order to address whether patterns of tree seedling composition established shortly after fire persist later into succession, we (1) examined changes in patterns of tree density and biomass that occurred during two decades of post-fire succession, and (2) compared patterns of tree composition in relation to variations in fire severity in four burned black spruce forests in interior Alaska after 10-20 years of succession. We found that initial effects of fire severity on recruitment and establishment of aspen and black spruce were maintained later in succession. Our results suggest that severe burning will lead to a switch from black spruce to deciduous canopy dominance. Thus, the assessment of fire severity, a parameter that can be accurately measured on the ground, and with increasing reliability by remote sensing, is a powerful method to predict habitat characteristics such as species composition, stand structure and productivity. This insight can be used by managers to estimate future forage resources and proactively implement conservation measures commensurate with these scenarios.

9:25 am

IMPACTS OF SHORTENED FIRE RETURN INTERVALS ON VEGETATION AND WILDLIFE HABITAT

Jennifer Barnes

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Understanding the impacts of shortened fire return intervals may be important to understand the impacts of climate warming on vegetation and wildlife habitat. In general spruce forests in interior Alaska have a fire return interval ranging from 60-150 years. Yet 13% of the area burned in Alaska in the past 60 years has burned more than once, including several long-term monitoring plots established by the National Park Service fire ecology program. Ten plots in Denali National Park and Preserve and four plots in Yukon-Charley Rivers National Preserve that burned in the mid-1980s were burned again 15-19 years later. This coincidence provided a unique opportunity to investigate what characteristics predisposed these sites to burn again and to track vegetation response to multiple fire events at varying levels of severity. Preliminary analyses suggest that multiple fire events at high levels of burn severity can lead to a forest community type conversion not evident at lower levels of severity. For example, a black spruce forest burned twice at a high level of severity is now a grass-land with a few deciduous trees with little resemblance to the pre-fire community. Yet some forest types appear to be fairly resilient to multiple burns - particularly aspen/spruce forests and black spruce/tussock forests.

10:10 am

MANAGING MOOSE POPULATIONS FOR HARVEST IN WILDLAND FIRES

Kalin Kellie Seaton

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Wildfire can rejuvenate habitat, creating the early-successional forage that is favored by several species of wildlife, including moose (*Alces alces*). Wildlife managers indirectly influence the location and occurrence of wildfire through cooperation with land managers in setting fire management options. This pre-fire initiative by managers may increase size of wildfires through reduced suppression, but forecasting functional and numeric response by moose depends on specific post-fire protocols for burn evaluation and active management of moose populations. Several post-fire factors must align for wildfire to result in both higher moose density and additional harvest by hunters. These factors can be grouped into 3 categories: burn characteristics, population dynamics and harvest limitations. Further, for some factors, additional research is needed to quantify their effect on moose dynamics and harvest. I propose an eventual framework for post-burn management of moose that includes several key elements: (1) prioritization of recent burns for additional evaluation, (2) field protocols for evaluating moose habitat in high-priority burns, (3) options for monitoring and active management of wildlife during burn regeneration, (4) cooperative or administrative actions needed to increase hunt participation, and (5) a timeline for implementation.

10:35 am

CONSIDERATIONS IN MANAGING WILDLAND AND PRESCRIBED FIRE FOR RESOURCE OBJECTIVES

Tom Kurth

State of Alaska, Division of Forestry. Contact: tom.kurth@alaska.gov

Given the legal, logistic, and sociological complexity of fire and fuels management near settlements, what are the possibilities and/or expectations for using fuels mitigation and prescribed or wildland fire to shape landscape processes for wildlife habitat objectives? An historical perspective and options for prescribed fire, wildland fire, or mechanical disturbance for hazard fuels reduction and/or habitat enhancement within Alaska. Included will be a look at the Alaska Interagency Wildland Fire Management Plan, prescribed fire projects, hazard fuels reductions and the associated implications.

Forestry Session

1:20 pm

ASSESSMENT OF WOODY BIOMASS TO EVALUATE SUPPLY OPTIONS FOR HEAT AND POWER COGENERATION

Douglas Hanson*

State of Alaska, Department of Natural Resources, Division of Forestry.

Contact: douglas.hanson@alaska.gov

With Alaska's dependence on expensive diesel as a primary heating fuel source, many communities are experiencing severe fiscal impacts. This situation has prompted an intense study of lower cost options which include the use of woody biomass, especially in the Interior, where the extreme cold winters serve to greatly exasperate the problem. Within the Tanana Valley a forest inventory update on state forest lands is mostly complete and is being used to evaluate the merchantable supply of biomass by forest stand type. Typically forest inventories focus on the merchantable portion of the tree stem that is generally to a minimum top of 4-inches or greater. By utilizing entire tree weights of green trees sampled in the Tok area, the inventory is able to categorize an above ground green mass biomass estimate as well. These measurements as well as other site data collected during the inventory field work allow estimations of a broad range of stand type characteristics useful for estimating biomass supply over a broad area. The inventory is discussed in terms of providing data for sustainably producing biomass as well as exploring further information valuable to better describe the merchantable supply.

1:45 pm

CHALLENGES AND OPPORTUNITIES WITH BIOMASS REGENERATION FOR HARDWOODS AND SOFTWOODS IN THE BOREAL FOREST

Clare Doig

Forest and Land Management, Inc. Contact: cdoig@forestandlandmanagement.com

This will be a discussion of the problems encountered with regeneration, particularly issues of competing vegetation as well as small and large animal predation. Examples of regeneration failures and successes in areas ranging from the coastal region to interior Alaska will be described. Potential benefits of favoring regeneration of specific species will also be discussed. Suggestions for investigation, trials, and research will be offered.

* - student

2:10 pm

CONSIDERATION OF SOIL NUTRIENT SUPPLY IN CHOOSING TREE HARVEST AND PROCESSING STRATEGIES

John Yarie

Forest Sciences Department, SNRAS, University of Alaska Fairbanks Contact: jayarie@alaska.edu

One consideration when planning a forest harvest is what effect the harvest will have on the subsequent ecosystem dynamics. A common concern is how much of the sites nutrient capital is being removed at time of harvest. This could be a significant amount when a biomass harvest is performed and all the aboveground woody material is removed or when a prescribed burn is planned after a harvest. In addition to nutrient availability there are a number of factors that can affect subsequent forest growth. These can be recognized through use of the state factor concept. The factors will be related to site climate, soils present and the actual physiographic setting of the site. A key method to examine potential nutrient limitations is through the use of fertilization experiments. A number of fertilization studies performed in interior Alaska indicate that site nutrient capital may not be the primary limiting factor. Results from simple thinning and moisture limitation studies indicate that the primary limiting factor of forest growth may be water.

2:35 pm

EFFECTS OF CLIMATE CHANGE ON ALASKA BOREAL FOREST GROWTH, HEALTH, AND REGENERATION

Glenn Juday¹, Claire Alix², and Tom Grant¹

¹University of Alaska Fairbanks, School of Natural Resources and Agricultural Sciences and ²University of Paris Sorbonne. Contact: gpjuday@alaska.edu

In Interior Alaska white and black spruce, aspen, and Alaska birch on low elevation sites register a strong negative radial growth relationship to summer temperatures, in particular May and July mean temperatures. Precipitation, particularly in late winter and midsummer, is a supplemental positive factor in growth. Floodplain white spruce trees along major Interior rivers transition from negative temperature responses in Interior Alaska to positive in western Alaska near tree limit. White spruce at mountain treeline sites display both negative growth response to July temperature and positive response to spring temperatures, with negative response dominant in the east and positive response dominant in the west. Across boreal Alaska summer temperatures increased abruptly in 1974 and have remained historically high since, with resulting climatic favorability for Interior tree growth on low elevation sites at extreme low levels since then. Defoliating and wood boring insects, including aspen leaf miner, spruce engraver beetle, and spruce budworm, reached outbreak population levels across boreal Alaska in the past 3 decades, partly from release of direct temperature control on the insects and partly from increased tree susceptibility. Temperatures and precipitation on many Interior sites are now beyond values that historically defined distribution limits for Interior tree species. Over the past 25 years tree mortality of 50% or more occurred in several long-term monitoring plots in mature stands on productive sites. Wildlife dependent on older forest have experienced habitat loss and are risk for further loss. To date trees have successfully regenerated on most disturbed sites.

3:20 pm

MODELING FOREST LANDSCAPE DYNAMICS TO EVALUATE MANAGEMENT OPTIONS

Nancy Fresco and T. Scott Rupp

University of Alaska Fairbanks SNAP Program. Contact: nlfresco@alaska.edu

Climate change is driving complex landscape processes across interior Alaska forests, including permafrost thaw, alterations in fire cycles, and biome shifts. Each of these phenomena is likely to impact forest health, succession, composition, and biomass, both independently and in conjunction with one another. The Scenarios Network for Alaska and Arctic Planning (SNAP) is a research group within the International Arctic Research Center (IARC) at UAF, focused on modeling climate change impacts and interpreting these models, in order to provide useful planning tools for a wide range of stakeholders via collaborative projects. Scientists, programmers, GIS experts, and outreach specialists at SNAP rely on a selected set of Global Circulation Models (GCMs), state-of-the-art model downscaling techniques, linked models, published data and information, and scenarios planning methods. SNAP provides maps, data, and clear explanations of all methods and assumptions, in order to help inform land managers and assist in planning for an uncertain future. These tools can help guide the future of forest management in the boreal region.

3:45 pm

LONG TERM CONTRACTS FOR TIMBER SUPPLY FROM STATE LANDS --- HOW WE GET THERE

Michael Curran

Alaska Division of Forestry. Contact: michael.curran@alaska.gov

The Alaska Division of Forestry has received multiple requests for large scale, long term timber sales on state land for purchasers to produce wood based fuel for heat and electricity. Questions have risen regarding economic feasibility of industrial proposals, timber harvest operability, multiple use of public lands, sustainable use of renewable resources, reforestation standards, and the effect these large sales will have on the landscape over time.

My presentation attempts to address some of these questions by providing a better understanding of the Alaska Statutes that govern Division of Forestry's management and responsibilities on State forests, specifically related to multiple use, sustained yield, and reforestation. It is also important to understand the step by step process required to sell state timber, including informing and involving the public on each step. The decision to sell timber is based on public input and must be in the best interest of the State.

I conclude by noting that the Division of Forestry and the Alaska Department of Fish and Game need to collaborate on proposed state timber sales at the beginning of the process, so that timber sales can be developed that best fit the responsibilities of each agency.

Friday, April 5

CONTRIBUTED ORAL PRESENTATIONS (8:00 am - 5:00 pm)

8:00 am

ASSESSING THE DISTRIBUTION AND GENETIC DIVERSITY OF THE ALASKAN HARE

Michelle Cason^{*1}, Link Olson², and Travis Booms³

¹University of Alaska Fairbanks, ²University of Alaska Museum, and ³Alaska Department of Fish and Game. Contact: mmcason@alaska.edu

The Alaskan hare (*Lepus othus*) is an important and long-utilized subsistence and sport-hunted mammal endemic to western Alaska for which there is no closed season or bag limit. Harvest levels, conservation status, distribution, and genetic diversity within *L. othus* have not been evaluated. The species has received very little research or management attention and even basic information such as its geographic distribution is inconsistent among the few published accounts, particularly with regard to its occurrence north of the Brooks Range. This study aims to exhaustively review and update the distribution of *L. othus* and examine its phylogeographic structure. To do this, we are georeferencing *L. othus* specimens from museums across North America and collecting additional voucher specimens from poorly-sampled regions in western Alaska. Using nuclear and mitochondrial DNA from these samples and from museum specimens, we will examine the extent of gene flow within *L. othus* throughout its range. This research will provide important and novel information useful for better understanding and managing this unique Alaskan endemic.

8:20 am

COMPARISON OF WILDLIFE RESOURCE SELECTION FUNCTION METHODOLOGIES: A CASE STUDY INVOLVING MOUNTAIN GOATS IN SOUTHEASTERN ALASKA

Dave Gregovich, Rod Flynn, and Kevin White

Alaska Department of Fish and Game, Division of Wildlife Conservation.
Contact: dave.gregovich@alaska.gov

The advent of GPS technology has resulted in an unprecedented amount of location data for wildlife populations. Resource selection functions (RSFs) implemented via generalized linear models (GLM) have received broad use for modeling habitat selection informed by such GPS locations. However, multiple RSF approaches have been published in the scientific literature, and there is no consensus on which methodology provides the most robust and reliable results. Furthermore, the consequences of methodology choice with regard to ecological and management conclusions have not been thoroughly investigated to date. Here we compare and contrast four approaches-pooled GLM, two-stage GLM, mixed-effects GLM, and generalized estimating equations (GEE)-using data collected during winter from mountain goats (n=47) fitted with GPS-collars (mean number of locations = 438, range 301-534) in southeastern Alaska. Specifically, we compare coefficient and variance estimates, top selected models, output RSF surfaces, and cross-validation performance to examine how analytical results and biological interpretations differ with respect to the methodology used. The management implications of model selection are assessed via differences in how a proposed road intersects goat habitats within the highest quartile of the RSF surface output from each respective modeling method. Overall, our analyses provide guidance for biologists interested in analyzing GPS telemetry data sets to model resource selection patterns of wildlife species for conservation and management purposes.

8:40 am

DISTRIBUTION AND MIGRATION OF AN IMPORTANT ARCTIC HERBIVORE, WITH IMPLICATIONS FOR SHRUB EXPANSION

Katie Christie*, Mark Lindberg, Roger Ruess, and Joel Schmutz

University of Alaska Fairbanks. Contact: kschristie@alaska.edu

Rock and willow ptarmigan are abundant arctic herbivores that may play an important ecological role in the Arctic by influencing the growth and architecture of expanding shrub communities. However, the extent and strength of ptarmigan browsing is unclear because their distribution and movement patterns are not well known, particularly in northern Alaska. The objectives of this study were to determine a) patterns of occupancy with respect to shrub cover, snow-free ground, and latitude, b) the extent of migration to northern Alaska in the spring, and c) differences in ptarmigan distribution between a high snow year and a low snow year. Aerial surveys were conducted from March to May in 2011 and April to May 2012 in northeastern Alaska. In areas with at least 40% shrub cover, the probability of occupancy of ptarmigan or their tracks was >0.90 , indicating that patches of shrubs associated with riparian areas had a high probability of being visited (and browsed) by ptarmigan. Occupancy increased throughout the spring in both years, indicating that ptarmigan undergo a spring migration from southern wintering areas to breeding areas north of the Brooks Range. Occupancy decreased with increasing latitude early, but not late in the spring, again indicating a pattern of seasonal northward movement. Ptarmigan distribution and migration in the Arctic are linked to expanding shrub communities on a wide geographic scale and these relationships may be shaping both rates and patterns of shrub expansion and ptarmigan population dynamics.

9:00 am

BROWSE INTAKES OF CARIBOU DURING SUMMER

Daniel Thompson* and Perry Barboza

Institute of Arctic Biology, University of Alaska, Fairbanks. Contact: dan.thompson2@alaska.gov

Shrubs are becoming more prevalent in subarctic and arctic ranges used by caribou (*Rangifer tarandus*), but we do not know if caribou can tolerate increasing proportions of shrubs in their mixed summer diets. We measured instantaneous intakes of resin birch (*Betula glandulosa*) and feltleaf willow (*Salix alaxensis*) by caribou over the summer. We also measured daily intake of a formulated diet to determine that caribou increased food intakes (64.1 to 86.7 g DM/kg^{0.75}/d) while animals gained body mass (96.8 to 113.5 kg) over summer. We estimated that caribou required 645 kJ/kg^{0.75}/d of energy intake to maintain body mass and 1113 kJ/kg^{0.75}/d to gain body mass for autumn reproduction. Caribou had the same bite mass (9.7 mg/bite/kg^{0.75}) and instantaneous intake rate (0.17 g DM/min/kg^{0.75}) on both forages even though resin birch contained more phenols (64.2 vs. 32.5%) and less available protein (6.2 vs. 10.2%) than feltleaf willow. We estimated that if a 100-kg female caribou can sustain these high rates of intake, it would need to consume 2.4 to 8.7 kg of fresh willow or birch with a minimum eating time of 3.1 to 8.5 h/day to meet energy maintenance and mass gain requirements. However, resin birch is unlikely to provide enough N to meet demands for maintaining body protein or mass gain. Carrying capacity for caribou in an increasingly shrubby arctic may depend on abundance and diversity of plants as caribou may need to consume other forages to offset high toxin loads and low protein intakes from shrubs.

9:20 am

DENSITY-DEPENDENT EFFECTS ON BODY CONDITION AND REPRODUCTION IN COASTAL ALASKA MOOSE POPULATION

Kevin White, Neil Barten, Stacy Crouse, and John Crouse

Alaska Department of Fish and Game, Division of Wildlife Conservation. Contact: kevin.white@alaska.gov

Population density can exert strong effects on nutritional and life-history characteristics of animal populations. In coastal Alaska, moose (*Alces alces*) populations have recently colonized highly productive post-glacial landscapes resulting in irruptions of local moose populations and conservation concerns related to habitat degradation and sustainability. During 2003-2010, we concurrently monitored moose density, body condition and reproductive parameters to characterize population-level effects of experimental reductions of a free-ranging moose population on the Gustavus forelands in coastal Alaska. During this period the Gustavus moose population declined from extremely high (5.5 moose/km²) to moderate (2.5 moose/km²) densities. Radio-collared adult female moose (n = 73) were captured each fall (n total = 147) and spring (n total = 192) to characterize longitudinal changes in body condition and reproductive performance in relation to experimental reductions in moose density. Overall, we detected a significant inverse linear relationship between moose population density and body condition (fall, $r^2 = 0.94$; spring, $r^2 = 0.42$), pregnancy ($r^2 = 0.74$), twinning ($r^2 = 0.52$) and fecundity ($r^2 = 0.62$); occurrence of reproductive pauses was positively related to population density ($r^2 = 0.78$). In addition, we documented a significant nutritional cost of reproduction that persisted across a range of population densities. Specifically, individuals with a calf a heel had significantly lower fat reserves than barren cows irrespective of population density. Overall, this study provides a compelling empirical example of how moose nutritional condition and reproduction are influenced by changes in population density.

9:40 am

MOOSE HUNTER SURVEY ON ANTLERLESS HUNTS

Todd Brinkman and Peter Fix

University of Alaska Fairbanks. Contact: tjbrinkman@alaska.edu

Moose (*Alces alces*) population levels in some areas of Interior Alaska have been above Alaska Department of Fish and Game's population goals since 1999. According to biological indicators used by ADF&G, moose in these areas are nutritionally stressed. To increase hunting opportunities and return the population to a more ecologically sustainable level, antlerless hunts were liberalized in 2004. Since then, antlerless hunts have become increasingly controversial and divisive among hunters and between biologists and hunters. We surveyed 2,500 Alaska residents who hunt moose in Interior Alaska to collect information on antlerless hunts. Our goal was to assess general attitudes toward antlerless hunts and to understand how the context of hunts influences acceptability. We received 863 completed surveys. Approximately 13%, 72%, and 15% of hunters thought antlerless hunts were always unacceptable, acceptable under certain situations, and always acceptable, respectively. In our paper, we also describe how hunter perceptions of antlerless hunts change with location, timing, and type of hunt along with several individual hunter characteristics (e.g., harvest success, trust in ADF&G data).

10:20 am

LOW DENSITY WOLF AND MOOSE POPULATIONS ON THE YUKON FLATS OF ALASKA: INSIGHTS INTO PREDATION FROM THE PREDATOR'S AND PREY'S PERSPECTIVE

Bryce Lake¹, Mark Bertram¹, Nikki Guldager¹, Jason Caikoski², and Robert Stephenson²

¹US Fish and Wildlife Service - Yukon Flats National Wildlife Refuge and ²Alaska Department of Fish and Game. Contact: Bryce_Lake@fws.gov

Moose density, wolf density, wolf kill rate, and wolf predation rate on moose were examined during 2009-2010 on the Yukon Flats of eastern Interior Alaska. Among regions where wolf kill and predation rate have previously been studied, the Yukon Flats is unique because moose are the sole ungulate prey, and both moose and wolves occurred at densities among the lowest in North America. Moose density was estimated with a geospatial population estimation technique. Wolf density was estimated by counting wolves in packs and determining the population area occupied by packs. Kills of moose were located using VHF and GPS technology, and by following tracks and flying to GPS location clusters. Moose (0.05-0.13 moose/km²) and wolf densities (3.5-3.7 wolves/1000km²) were low. Wolf kill rate estimates (mean, 95% CI) were highest in November (0.033 moose/wolf/day, 0.011 - 0.055) and lowest in February (0.011, 0.002 - 0.02). Predation rate was 12% during 151 days of winter (November 2009-March 2010). Kill rates at low moose density were similar to those from other populations where moose occurred at higher densities. We suggest that few, if any, low density systems in Alaska and the Yukon experience a density-dependent phase in the functional response, and instead that wolves respond numerically to changes in moose density or availability. Through numerical response, wolf predation rates may approximate the annual growth potential of the moose population, contributing to persistent low densities of moose and wolves on the Yukon Flats.

10:40 am

RELATIONSHIP OF COMMUNITY CHARACTERISTICS TO HARVEST REPORTS BY ALASKAN MOOSE HUNTERS ABSTRACT

Jennifer Schmidt, and Terry Chapin

University of Alaska Fairbanks. Contact: jjschmidt@alaska.edu

In Alaska, wildlife managers often estimate harvest from harvest tickets returned by hunters. We evaluated the influence of community characteristics (e.g., population size, ethnicity) on harvest report rates for moose. Reporting rate was defined as the number of moose reported harvested from harvest tickets versus household surveys, assuming the latter are more likely to be accurate. Alaskan communities reported a median of 27% via harvest tickets versus subsistence surveys and this did not change over the 23 years studied. Regression and t-tests showed that harvest-report ratio was more strongly influenced by cultural factors than by recent western influences. The percentage of the community that was indigenous, used and shared moose, and absence of a wildlife biologist or road access were associated with low harvest-report ratios. We suggest reporting rates could be improved by placing more emphasis on the benefits of reporting harvests and increasing two-way dialogue between hunters and managers.

11:00 am

RELATIONSHIPS BETWEEN MOVEMENT, DIET, AND HABITAT QUALITY OF URBAN MOOSE

Joseph Welch¹, Sean Farley², Perry Barboza¹, Don Spalinger³, and David Battle²

¹University of Alaska Fairbanks, ²Alaska Department of Fish and Games, and ³University of Alaska Anchorage. Contact: jwelch@abrinc.com

We studied movements and foraging behavior of adult female moose in urban Anchorage and Joint Base Elmendorf-Richardson, Alaska. From summer to winter, moose reduced the proportion of time spent active from 52% (12% SD) to 40% (6% SD), reduced bedding events from 7.1 (0.9 SD) to 6.0 (1.1 SD) per day, and reduced mean turning angles between locations from 87° (18° SD) to 70° (15° SD). Mean distance traveled between locations did not change by season. Moose diet composition changed from summer to winter with major differences in diets explained by the increase in the proportion of *Betula papyrifera* and the decrease in the proportions of minor shrub components. Moose were more active and more likely to forage in higher quality habitats when foraging on more digestible diets that contained less *B. papyrifera* and more *Salix* spp. In winter, when energetic costs are high and forage quality is relatively low, moose in our study area reduced movements and sought higher quality diets and habitats to increase energetic returns. Habitat improvement projects should consider the importance of willow and forbs and locate improvements in areas that will minimize conflicts with people.

11:20 am

FINE-SCALE STRUCTURE OF CALVING LOCATIONS FROM NATAL PHILOPATRY IN MOOSE

Kevin Colson^{*1}, Kevin White², and Kris Hundertmark¹

¹University of Alaska Fairbanks and ²Alaska Department of Fish and Game. Contact: kcolson@alaska.edu

Female natal philopatry has often been implicated as an important factor in moose (*Alces alces*) home range formation, with many populations showing behavioral evidence of sympatric home ranges among related individuals. However, observed rates of dispersal often disagree with the greater observed rates of population expansion, and several genetic studies have failed to detect regional genetic structure expected from philopatry. Here we examine data from radio-marked individuals (n = 100) collected in two separate populations in Berners Bay and Gustavus, Alaska to examine the extent to which moose calving locations are structured by relatedness. Both populations are significantly more related to individuals within the first 1.5 km (mean $r = 0.030$), with Berners Bay individuals also being significantly more related to individuals between 1.5 km and 3 km away ($r = 0.054$). When both populations were combined, the overall pattern of spatial relatedness was significant ($P = 0.007$), with mean $r = 0.030$ between 0 km and 1.5 km ($P = 0.046$) and $r = 0.023$ between 1.5 km and 3 km ($P = 0.044$). Additionally, 11 % of the population shows significantly positive relatedness to their five nearest neighbors, with an overall average $r = 0.14$ of those focal individuals to their neighbors. Habitat, barriers to dispersal, population density and harvest strategy may be factors in promoting the development of this structure. This study presents the first known evidence for fine-scale genetic structure in moose and natal philopatry to calving locations in moose.

11:40 am

DIETARY DIFFERENCES OF NORTHERN RED-BACKED VOLES (*MYODES RUTILUS*) FOUND ALONG AN ALASKAN MEGATRANSECT: A STABLE ISOTOPE ($\delta^{15}\text{N}$ AND $\delta^{13}\text{C}$) STUDY

Andy Baltensperger^{*1}, Falk Huettmann², and Jeffrey Welker³

^{1,2}University of Alaska Fairbanks and ³University of Alaska Anchorage. Contact: abaltens@alaska.edu

The arctic and boreal biomes of the circumpolar North are undergoing dramatic changes in climate, geographic distribution, and ecosystem structure. For small mammals in Alaska, describing realized dietary niche space is one tool for identifying competition potential between co-occurring species. We sampled blood and hair from 37 Northern red-backed voles (*Myodes rutilus*) at four low-elevation sites along a latitudinal mega-transect from south-central Alaska to the North Slope. Stable isotope analyses ($\delta^{15}\text{N}$ and $\delta^{13}\text{C}$) of blood, hair, and potential food sources were used to quantify dietary preferences of *M. rutilus*. We used Bayesian isotopic mixing models to determine the range of dietary input proportions for five vegetation functional groups, and to plot the isotopic niche breadth of *M. rutilus* at each location. Machine learning methods were employed to create an explanatory model of isotopic variability among locations using 16 spatially explicit variables. We found high variation in proportions of dietary groups among locations and differences in hair and blood results provided new insight into the temporal structure of foraging patterns of *M. rutilus*. Important geographic explanatory variables included elevation, geology, and vegetation. The identification of competition potential within newly emerging species interactions will be of particular conservation concern for the persistence of small mammal species in the future.

1:40 pm

THE EFFECT OF RESOURCE DEVELOPMENT ON ARCTIC FOX SURVIVAL ON THE ALASKAN ARCTIC COASTAL PLAIN

Garrett Savory^{*1}, Christine Hunter¹, Brian Person², Neil Lehner¹, and Nathan Pamperin³

¹University of Alaska Fairbanks, Institute of Arctic Biology, ²North Slope Borough, Department of Wildlife Management, and ³Alaska Department of Fish and Game. Contact: gasavory@alaska.edu

Mesocarnivores adapted to life near humans often increase their fitness by taking advantage of resources from anthropogenic sources. In Prudhoe Bay (PB), Alaska, arctic fox (*Vulpes lagopus*) forage on anthropogenic foods at sites such as dumpsters or the landfill. It is not known whether this benefits fox survival. Our primary objective was to determine whether arctic fox survival was higher in a developed area, PB, than an undeveloped area, the National Petroleum Reserve-Alaska (NPR-A). Using satellite collar data from 2004, 2005, and 2009 we evaluated the effects of development, sex, season, distance traveled, and sea ice use on adult and juvenile fox survival. Two models, S(age+development+ice use) and S(age+development) had $\Delta\text{QAICc} < 1.0$. Both development and sea ice use had 2 times more support than any other factor. Without sea ice use, model-averaged estimates of annual survival (90% C.I.) were 0.40 (0.18-0.62) for NPR-A adults, 0.50 (0.31-0.69) for PB adults, 0.01 (0-0.04) for NPR-A juveniles, and 0.04 (0-0.08) for PB juveniles. Use of sea ice, observed only for NPR-A foxes, increased estimated survival rate. However, annual survival was still higher for PB than NPR-A adults unless NPR-A adults spent eight (seven for juveniles) or more 2-week periods on the ice over winter. The relatively small difference between PB and NPR-A fox survival rates suggests that access to anthropogenic foods has only a small survival benefit. Alternatively, there could be trade-offs between positive and negative effects of development, e.g. supplemental food and increased interspecific competition with red foxes (*Vulpes vulpes*), respectively.

2:00 pm

RELATIONSHIPS BETWEEN DIET, FECUNDITY, AND POPULATION ABUNDANCE AND COMPOSITION IN AMERICAN MARTEN IN INTERIOR ALASKA

Craig Gardner and Nathan Pamperin

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American marten (*Martes americana*) are the most valuable and harvested furbearer species in Interior Alaska. Marten abundance can vary greatly between years due to food availability, habitat changes, density dependent effects and overharvest. Mostly, due to expansive wilderness and relatively dispersed trapping patterns in the Interior, there are rarely any region wide management concerns. However, marten numbers can decline to low levels and depending on the fur market, the ability to forecast population trends becomes more important to managers and individual trappers. Little research has been conducted to identify indices to assess marten population status prior to, or during the trapping season. In Southeast Alaska, Flynn and Schumacher (2009) found that marten pregnancy rates were correlated with rodent abundance and there was a strong relationship between fecundity the previous year and current year numbers and mean age of harvested marten. We initiated a study to evaluate if monitoring gender, juvenile:adult female ratios, diet and fecundity (pregnancy rate and blastocysts counts) of harvested marten were useful in predicting marten population status and composition. During 2011 and 2012, we necropsied 308 and 1183 marten carcasses from 4 and 11 areas, respectively across the Interior and evaluated pregnancy rates and juvenile:adult female ratios. Dietary data analyses are pending. These data were adequate to identify the status of marten populations across the Interior and be used by trappers to structure their trapping efforts. We recommended trappers to cease their trapping efforts early during winter 2011-2012 and further reduce harvest this season (2012-2013).

2:20 pm

SPATIAL PATTERNS ASSESSING POSITIVE VERSUS NEGATIVE PERCEPTIONS OF HUMAN-BEAR ENCOUNTERS

Kim Jochum^{*1}, Andrew Kliskey², Kris Hundertmark¹, Dmitry Lisitsyn³, and Lillian Alessa²

¹Biology and Wildlife Department, University of Alaska Fairbanks, ²Biological Sciences, University of Alaska Anchorage, and ³Sakhalin Environment Watch. Contact: kajochum@alaska.edu

An important challenge for wildlife management is how to address an increasing human population encroaching upon wildlife habitat. People living close to wild lands, increasing access to wildlands via roads and motorized vehicles, competing resource demands (e.g. fish, berries), and recreational activities (e.g. hiking, fishing) are overwhelming current wildlife management techniques. Wildlife management must increasingly contend with the co-existence of people and wildlife in a diminishing wild. In recent years, the amount of human-wildlife conflicts and encounters has increased worldwide, and in many areas those interactions are perceived as negative. This study examines the distribution of positive versus negative and neutral perceptions of human-bear encounters across an urbanizing region in Alaska (Anchorage, Mat-Su, Kenai Peninsula), and in the Russian Far-East (South Sakhalin Island). Data were collected via semi-structured surveys throughout the study areas in summers of 2011 and 2012. Encounter locations were geo-referenced and analyzed using Geographic Information Systems. Kernel density estimators were used to generate density maps of encounter types. Results are currently preliminary, with an emphasis on density map comparison. Future significance testing of spatial correlations and modeling techniques ranking the importance of parameters will be discussed. Initial results show no obvious difference between spatially perceived positive versus negative encounters. These findings suggest that the major triggers of how human-bear encounters are perceived are independent of location, but rather dependent on the persons' individual perceptions towards encountering bears. Results will improve future human-bear encounter management through an increasingly positive and sustainable co-management of people and bears.

2:40 pm

DOCUMENTING PRESENCE OF WOLVERINES IN NORTHEAST OREGON AND IDENTIFYING INDIVIDUALS

Audrey Magoun and Patrick Valkenburg

Wildlife Research and Management (WRAM). Contact: amagoun@ptialaska.net

Wolverines are listed as a state endangered species in Oregon and have been proposed for listing under the federal Endangered Species Act. Until 2011, there had not been a documented record of a wolverine in Oregon for 17 years. In spring 2011, we documented the occurrence of 2 male wolverines and 1 wolverine of unknown sex in the Wallowa Mountains of northeastern Oregon, more than 60 miles west of known breeding range in Idaho. We also describe and illustrate a photographic method to identify individual wolverines and their breeding and lactation status.

3:20 pm

POPULATION STRUCTURE OF TWO RABIES HOSTS IN ALASKA

Elizabeth Goldsmith*, Christopher Clements, Elisabeth Himschoot, Ben Renshaw, Kris Hundertmark, and Karsten Hueffer

University of Alaska - Fairbanks. Contact: ewgoldsmith@gmail.com

Rabies virus is a zoonotic disease maintained in endemic infections in a variety of mammalian hosts across the globe. Researchers from the CDC identified three distinct variants of rabies virus localized to three regions of coastal Alaska (Kuzmin et al. 2008), whereas the Alaskan Interior and Southcentral are devoid of endemic rabies infection. Rabies is maintained as an endemic infection in Alaskan wildlife, primarily in the Arctic Fox population, and the presence of endemic rabies corresponds to the range of the Arctic Fox in Alaska. We hypothesize that the distribution of the three rabies variants in Alaska correlates with the population structure of Arctic and/or Red Foxes. Undergraduate students in the Hueffer lab at the University of Alaska - Fairbanks used mitochondrial DNA and microsatellite analysis to examine fox population genetics from DNA samples collected across the state. Preliminary analysis supports the hypothesis that there is decreased genetic mixing between Arctic Foxes in the three rabies regions in Alaska and that there is decreased genetic mixing in the Red Fox population between areas with endemic rabies and areas without endemic rabies. By understanding the population genetics of these two hosts of rabies in Alaska, public health officials and wildlife managers will be able better able to understand the impact of changing fox ranges due to development and climate change on rabies epidemiology across the state.

3:40 pm

SEXUAL SEGREGATION OF BROWN BEAR HABITAT USE ALONG THE SITUK RIVER

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The Situk River in Yakutat, Alaska is acclaimed for its productive salmon fisheries and abundant brown bear (*Ursus arctos*) population. Increasing angler use has raised concerns over conflicts between bears and people. We investigated the temporal and spatial resource use patterns of 8 brown bears (3 adult females, 3 adult males, and 2 sub-adult males) using GPS radiocollar data deployed between 2009 and 2012. We estimated brown bear probability density functions from locations within 500 meters of the stream (n=9,811) and generated kernel density estimators with smoothed cross validation bandwidth. We also calculated utilization distribution overlap indexes (UDOI) to evaluate the spatial separation between male and female brown bears. Habitat use of dominant vegetation types was proportional to available habitats in both sexes, though bears were found to be sexually segregated relative to concentrations of human activity. Male brown bear habitat use was concentrated further from the stream, roads, and trails. Female bears, with and without cubs, utilized habitats closer to trails and roads, supporting evidence of social segregation rather than habitat segregation. Adult male diurnal displacement from the river was pronounced as they were located twice as far from the river than females. Between late summer and autumn bear activity levels decreased and they remained closer to the river to consume salmon and fulfill energetic demands of hyperphasia. Activity sensor data showed female bears were active throughout the day with limited activity between 11pm and 5 am. Male temporal activity was bimodal with peak levels at 5 am and 7pm. Understanding brown bear habitat use patterns is important as agencies and public users cooperatively manage the recreational use of the area and plan for facilities development.

4:00 pm

ENHANCED RABIES SURVEILLANCE IN ALASKA USING THE DIRECT RAPID IMMUNOHISTOCHEMISTRY TEST (dRIT)

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Previously, rabies surveillance efforts in Alaska have been limited to evaluating intact brains of suspect animals due to behavioral observations or following potential exposure to a human or pet. In 2006, the Centers for Disease Control and Prevention (CDC) validated the direct rapid immunohistochemistry test (dRIT) using brain tissue unsuitable for standard techniques. While the direct fluorescent antibody test (dFA) remains the gold standard for rabies diagnosis, the dRIT has great potential for enhanced surveillance, as it is rapid and doesn't require fluorescent microscopy or opening the skull to retrieve appropriate samples. In 2011, CDC personnel trained and certified staff from agencies within Alaska to conduct dRIT on non-exposure specimens. We have utilized dRIT to test ~1100 brain specimens of free-ranging wildlife, representing 18 endemic species. Our surveillance efforts have identified rabies positive cases in 2.9% of red foxes (n=467, 403 via trappers) and 4.0% of arctic foxes (n=100) tested. Additionally, in 2012 we documented the first case of rabies in a wolverine in North America (2.4%, n=42). Results from dRIT for foxes are highly accurate and consistent with dFA; our broad investigation suggests occurrence of cross-reactions leading to false positive dRIT results for some other species (e.g. bats). Expanded surveillance has identified the significant risk of rabies transmission to resident trappers within the endemic region and detected presence in new species. Continuing to utilize the dRIT for passive and active rabies surveillance will enhance our ability to rapidly detect rabies outbreaks and changes in species or geographic distribution.

4:20 pm

MODELING AND MAPPING FOREST DIVERSITY OF INTERIOR ALASKA AT 1-KM² RESOLUTION FOR CURRENT AND POSSIBLE FUTURE CLIMATE CONDITIONS

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Proactive forest planning requires spatially accurate information about forest diversity. The most cost-efficient way to obtain this information is through modeling, i.e. predicting key forest diversity measures as a function of environmental factors. Patterns of forest diversity are less well known in the boreal forest of interior Alaska than in most ecosystems of North America. In order to understand the diversity patterns of this forest, we employed Random Forest analysis (machine learning) and the Boruta algorithm to predict tree species and tree size-class diversity for the entire region using a combination of forest inventory data and a suite of 28 predictors from public open-access data archives that included climatic, soil, distance, and topographic variables. We developed prediction maps for the current levels of tree size-class and species diversity and created maps showing the potential changes to these values under a null climate change scenario and the IPCC A1B "mid-range" scenario for the year 2050. The method employed here yielded good accuracy for the huge Alaskan landscape despite the exclusion of spectral reflectance data due to its transient nature. The results indicate that the geographic pattern of tree species diversity differs from the pattern of tree size-class diversity across this forest type and that future climate scenarios have different effects on tree species and tree size-class diversity depending on location. The results also suggest that human impact factors had a greater impact than the ecological factors in predicting the patterns of diversity within the boreal forest of interior Alaska.

4:40 pm

RESEARCH DESIGN FOR MANAGEMENT-RELATED WILDLIFE, HABITAT AND MONITORING FIELD STUDIES BASED ON MACHINE LEARNING ANALYSIS: NEW GOALS, OPPORTUNITIES AND WORKFLOWS

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Research design gets frequently ignored or overruled in field-based wildlife-habitat and monitoring studies. While this tends to compromise the overall outcome and statement of the project, other analysis methods like machine learning that can overcome such problems are still widely underused. However, they can allow for more flexibility and tend to be quite cost-effective reaching similar or better conservation management statements even. Here I elaborate on the “state” of optimized research design in wildlife studies (e.g. designed, haphazard or opportunistic), and how machine learning offers us a powerful contribution to overcome the earlier constraints and requirements such as the need for representative normally distributed and independent data free of gaps, autocorrelation and outliers. I provide our examples from the peer-reviewed literature for terrestrial applications (small mammals, diseases), as well as marine species (benthos and seabirds). Applications based on telemetry data (bears, caribou) are also discussed. I conclude with new and automated workflows (e.g. in R, Python) which allow to streamline the data collection, analysis and delivery for monitoring and conservation management projects opening up the in-time online delivery for complex wildlife research.

5:00 pm

BIRTH TIMING, SYNCHRONY, AND NEONATAL SURVIVAL IN A CRYPTIC FOREST UNGULATE: THE IMPORTANCE OF DETECTION METHOD FOR ESTIMATION

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In northern ecosystems, synchronicity of annual breeding and birth is common, with important consequences for reproductive success of adults and survival of young. Accurately estimating timing of births and subsequent survival of neonates is essential, yet remains challenging for many species. In large, highly mobile animals such as ungulates, the practice of using opportunistic captures of neonates to estimate neonatal mortality rates may result in significant underestimation of neonatal mortality, as many neonates die within hours or days of birth. We estimated neonatal survival and birth synchrony using two different capture methods, and examined effects of these alternative estimations on subsequent survival analysis. We used opportunistic neonate capture and capture using Vaginal Implant Transmitters (VITs) on estimation of reproductive success and synchrony in a population of Sitka black-tailed deer (*Odocoileus hemionus sitkensis*), a cryptic forest ungulate. We found that VIT and opportunistically captured fawns produced significantly different estimates of neonatal survival, mean birth date, and synchrony. VIT-based estimates indicated reduced survival during the first several days of life not found in the opportunistic sample, as well as both earlier and later births than detected opportunistically. We then examined the importance of synchrony to subsequent annual fawn survival, and found that early birth relative to the mean negatively affected neonatal survival, while relatively late births positively affected neonatal survival. Bears, the primary cause of mortality for fawns, may be prey switching during the later fawning period to salmon; this may combine with predator swamping during middle periods of fawning to produce the observed pattern. We conclude that capturing truly neonatal individuals is essential to accurate estimation of neonatal survival and birth timing and synchrony, and recommend the use of VITs for studies of ungulate reproduction to reduce bias in such estimates.

Notes